

Differential pair rules in gastrointestinal aGVHD

A visual catalogue of method checks, atlas-linked results, and new candidates

525 directional pair rules; 267 FOVs; clinical severity groups. Arrows describe conditional association, not causation.

Automatic severity selection ranks Control–Severe attraction and avoidance gaps separately. It requires at least 10 eligible FOVs at both endpoints and a visible change of at least 20 percentage points. Statistics remain available in notebook tables but do not gate this visual catalogue. Abundance appears only as a control.

1 Method proof

Recurring anatomical relationships lead the ranking, providing a first check of the method.

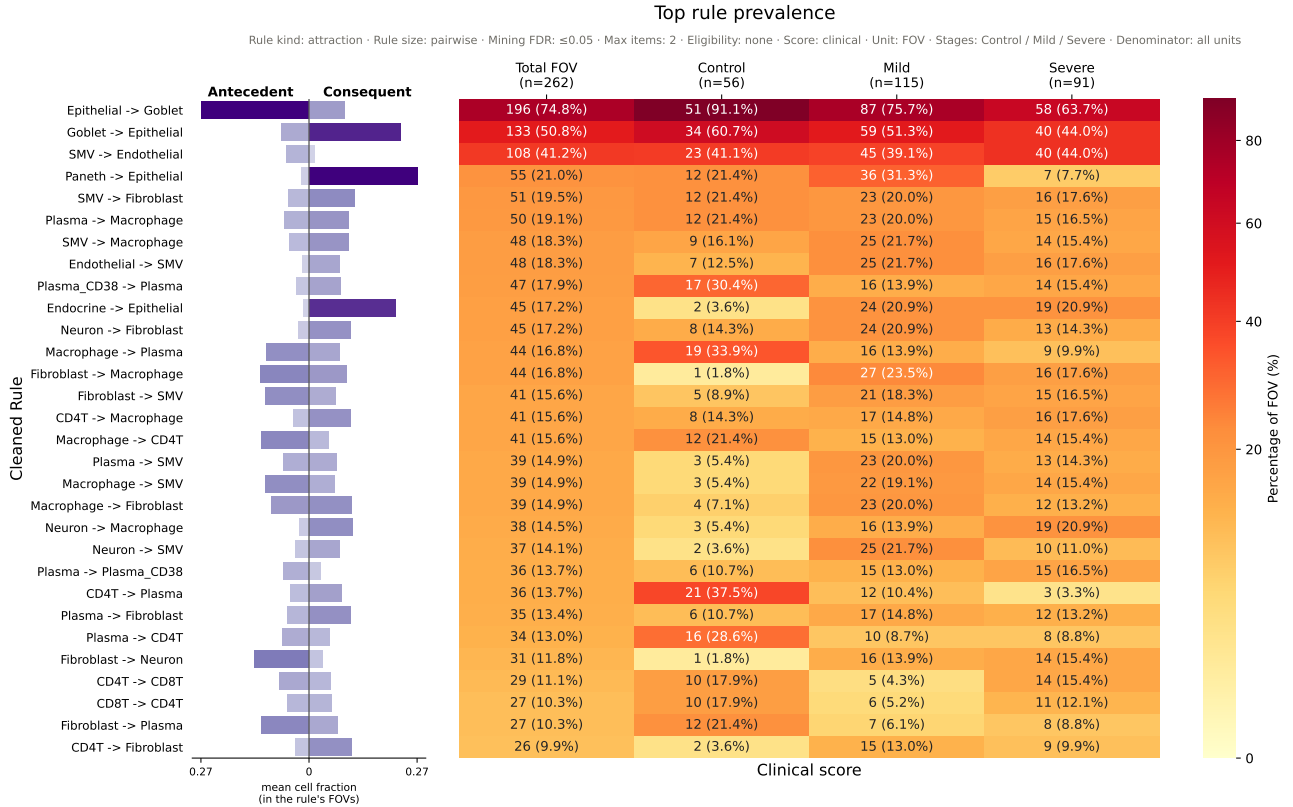


Figure 1: Rows are automatically ranked directional rules. Each cell gives the number and percentage of FOVs where the rule was mined, split by clinical stage and organ. This is an occurrence overview without the 20-cell eligibility control. It counts the 262 FOVs that carry a clinical score, not all 267.

The leading relationships also recur across patients, showing that they extend beyond a single biopsy.

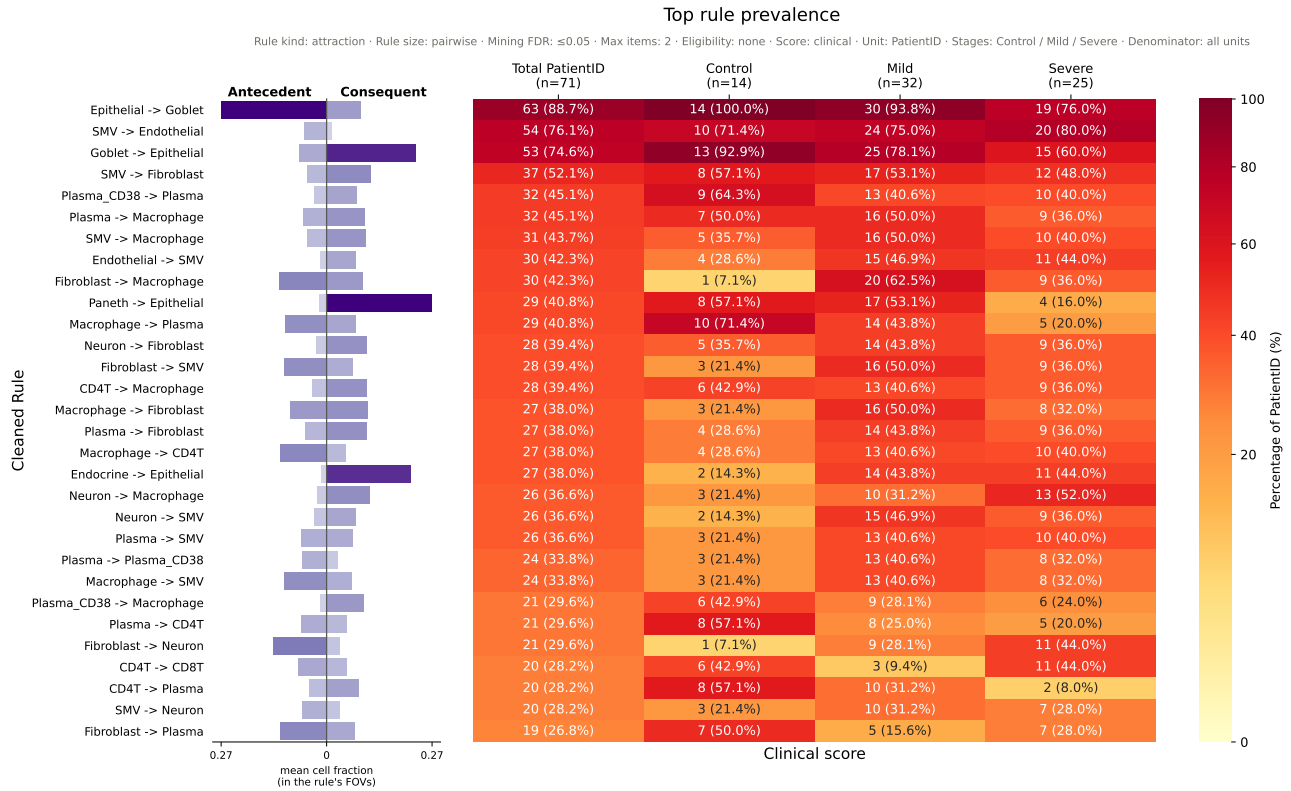


Figure 2: The calculation follows the preceding figure, but a rule counts once per patient when it appears in any of that patient's FOVs.

Epithelial → Goblet stays attractive in eligible fields, consistent with their shared intestinal architecture [2].

Colon · Epithelial → Goblet

Plots: FOV · Tests: biopsy · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs

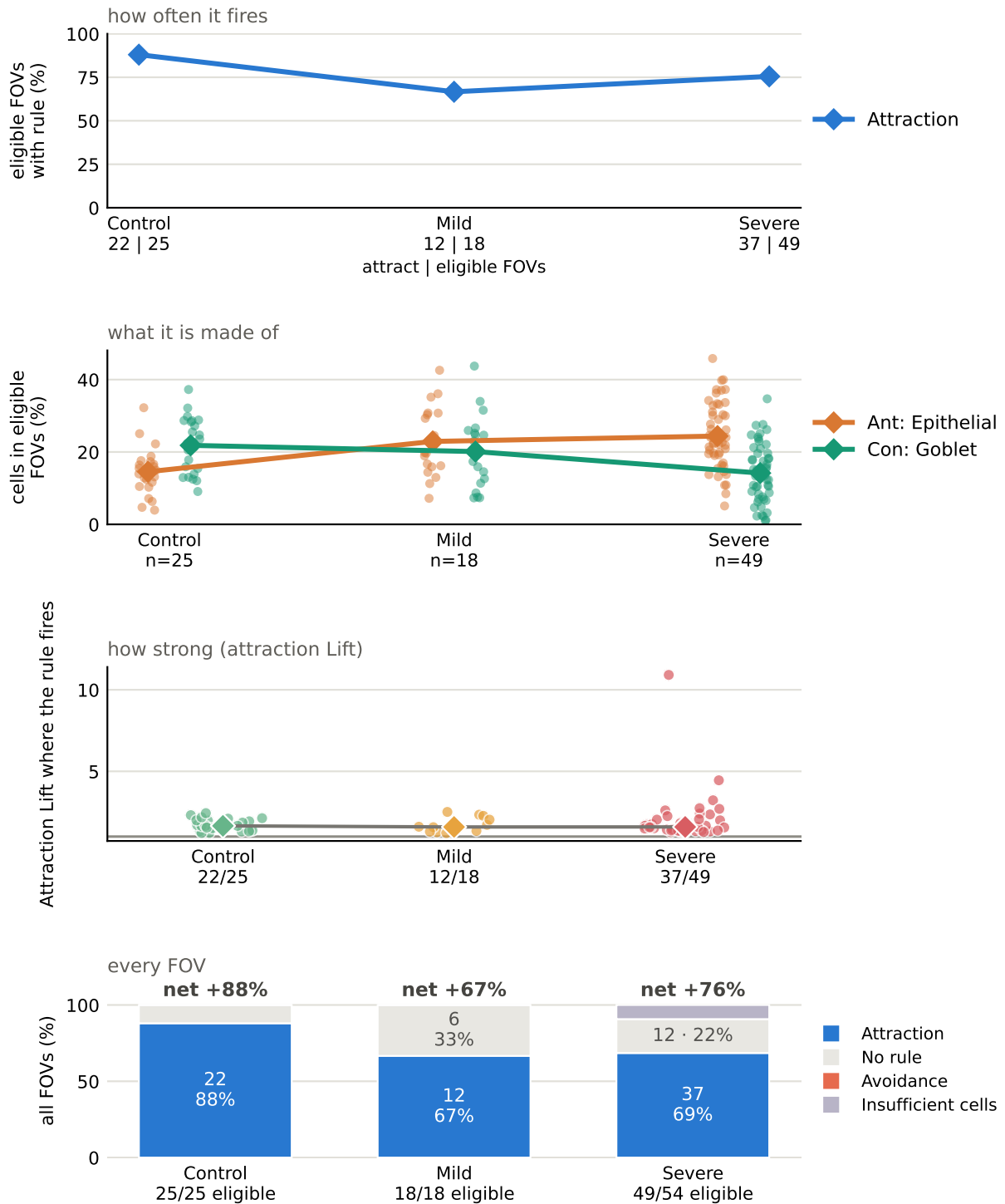


Figure 3: This is the first shared rule summary. The four rows use the same rule and clinical stages. **How often it fires:** attraction and avoidance are percentages of eligible FOVs; the numbers below each stage are raw attraction | avoidance | eligible counts, with an all-zero direction omitted. **What it is made of:** dots are the two cell-type shares inside each eligible FOV and diamonds are stage means; this is an abundance control, not a finding. **How strong:** dots are Lift in rule-bearing FOVs and diamonds are medians; Lift 1 is chance. **Every FOV:** bars include attraction, no rule, avoidance, and insufficient-cell fields, with net score above. An FOV is eligible only when it contains at least 20 cells of both rule types. Its state is attraction, avoidance, or no rule; net score is $100(N_{\text{attraction}} - N_{\text{avoidance}})/N_{\text{eligible}}$.

SMV → Endothelial stays attractive across stages, recovering a vascular relationship.

Colon · SMV → Endothelial

Plots: FOV · Tests: biopsy · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥20 cells/type · n = eligible FOVs

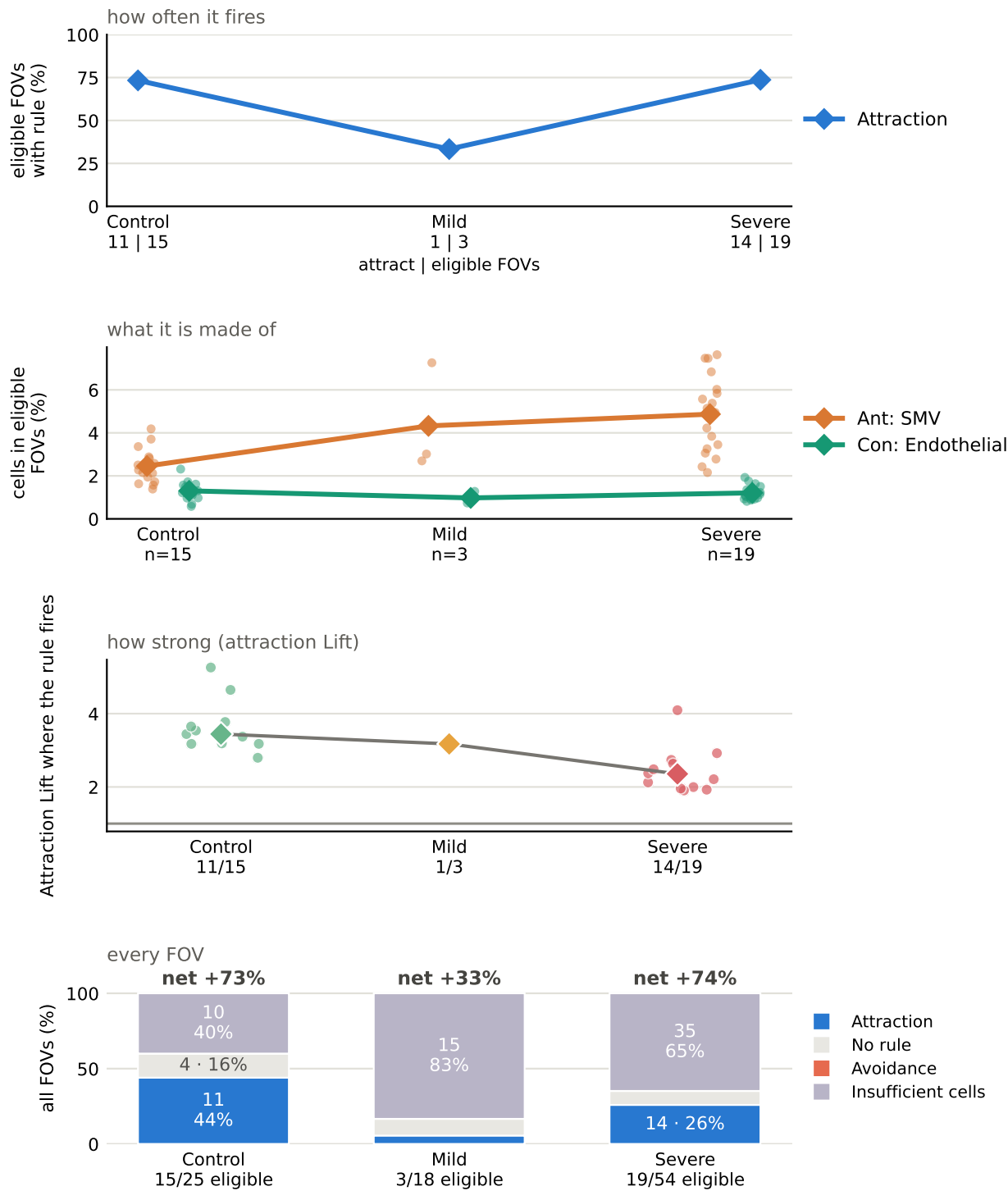


Figure 4: Read as in Figure 3.

Plasma_CD38 → Plasma stays strongly attractive, providing a check using related plasma-cell phenotypes.

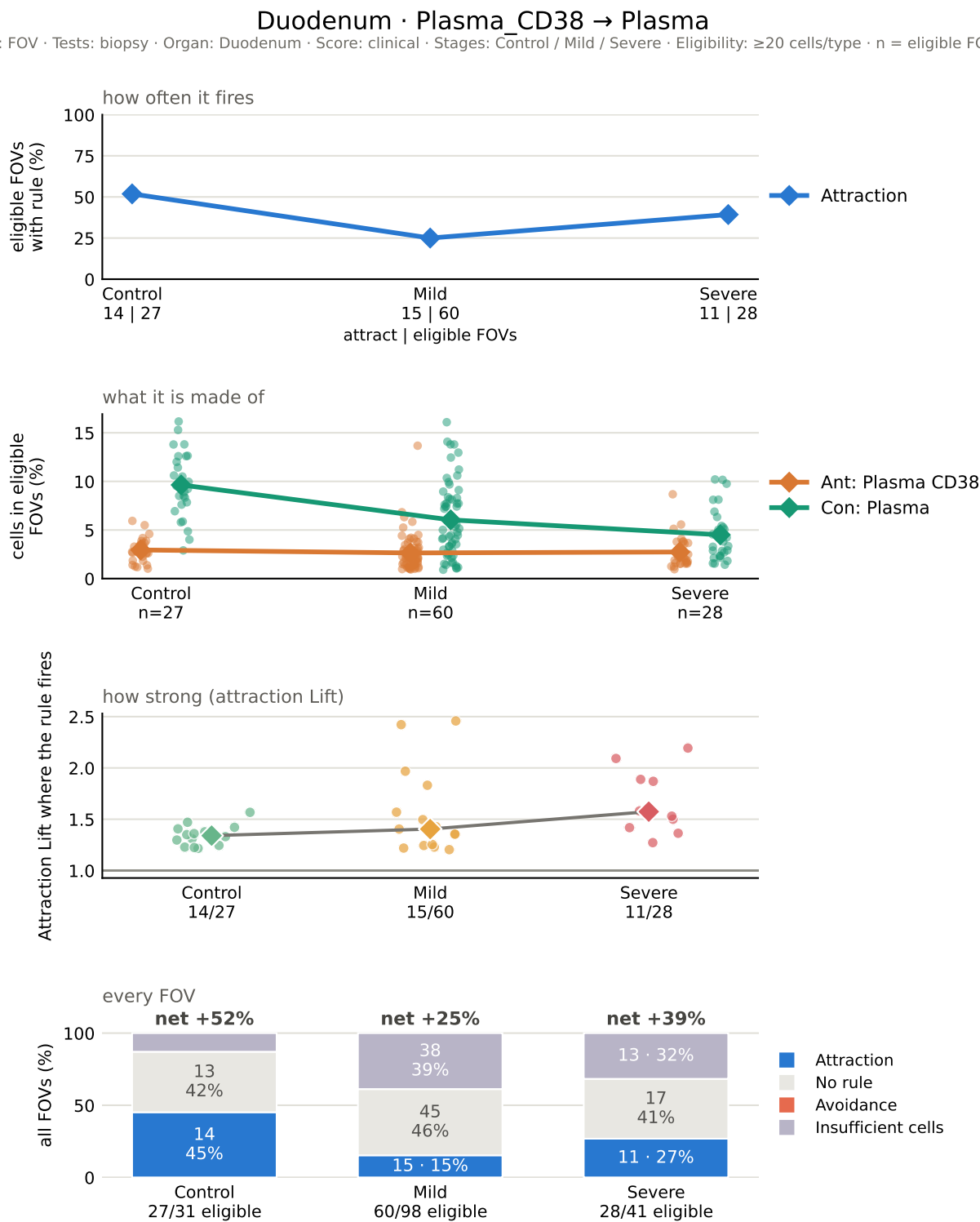


Figure 5: Read as in Figure 3.

1.1 Known biological pairs tested in both directions

Six literature-selected pairs are checked retrospectively; the notebook shows both directions, whole-organ prevalence, stage-specific results and typical tissue examples.

SMV → Endothelial occurs in about 70% of eligible colon fields, recovering an expected vessel-wall relationship [8].

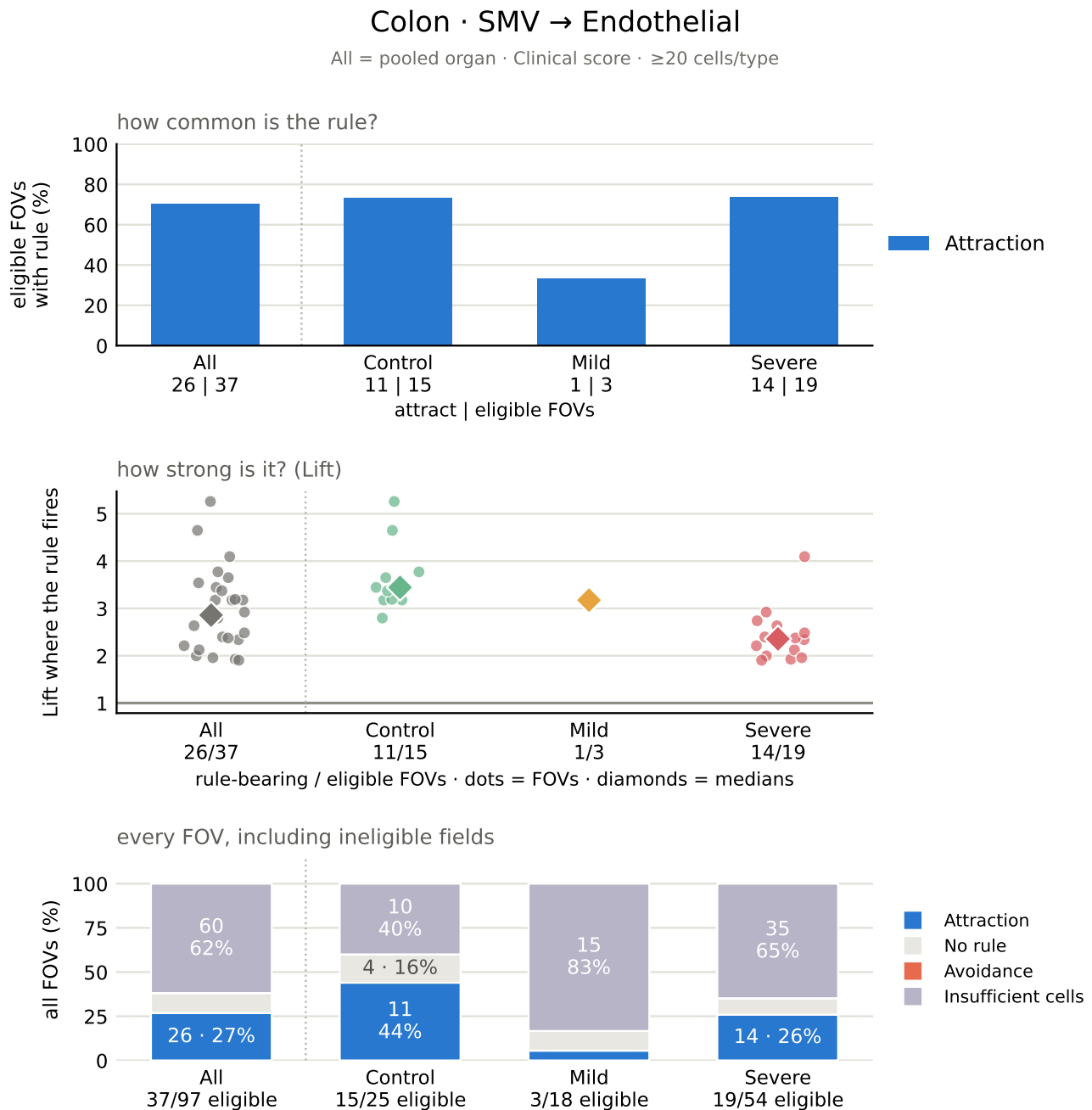


Figure 6: Both directions use the same FOVs with at least 20 cells of each type. All pools the organ's fields; the other columns split those same fields by clinical stage. Top: attraction and avoidance counts divided by eligible FOVs; counts below are attract | avoid | eligible, omitting an all-zero state. Middle: raw mined Lift in passing, eligible FOVs; dots are FOVs, diamonds medians, and Lift 1 is the independence baseline. Missing rules have no plotted Lift. Bottom: all FOVs, separated into attraction, avoidance, eligible without a rule, and insufficient cells. Counts below the top panel are attraction | avoidance | eligible FOVs. Here attraction occurs in 26/37 fields versus 12/37 in reverse.

The highlighted cells place the mined vascular relationship back into the tissue.

Colon SMV → Endothelial — representative FOVs

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: each observed state near median Lift; typical eligible no-rule FOV

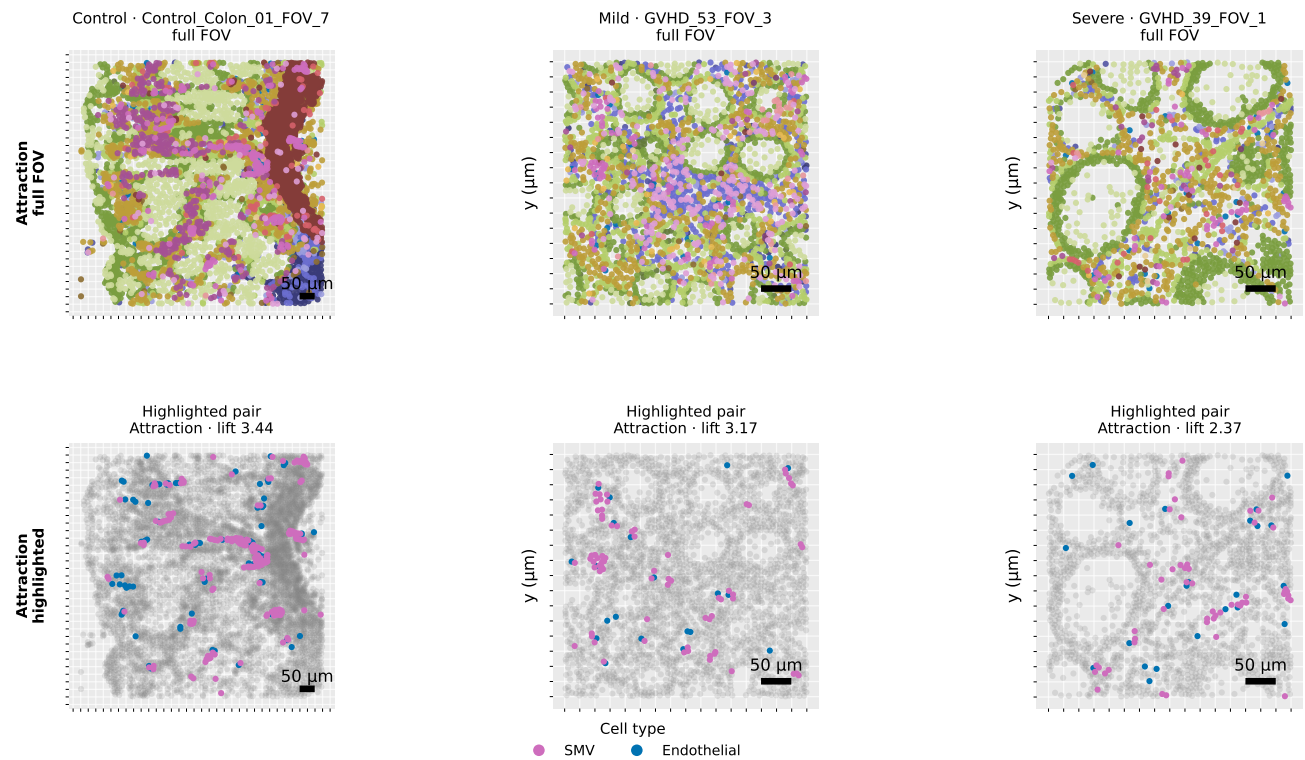


Figure 7: Each column shows a clinical stage, with the full FOV above and the same FOV highlighting only the two rule types below. One rule state is shown per figure. Maps are illustrative fields; highlighted cells are not individual detected interactions. Molecular signaling direction cannot be inferred from these maps.

SMV → Endothelial also recurs across 27 duodenal patients, extending the anatomical check beyond colon.

Duodenum · SMV → Endothelial

All = pooled organ · Clinical score · ≥ 20 cells/type

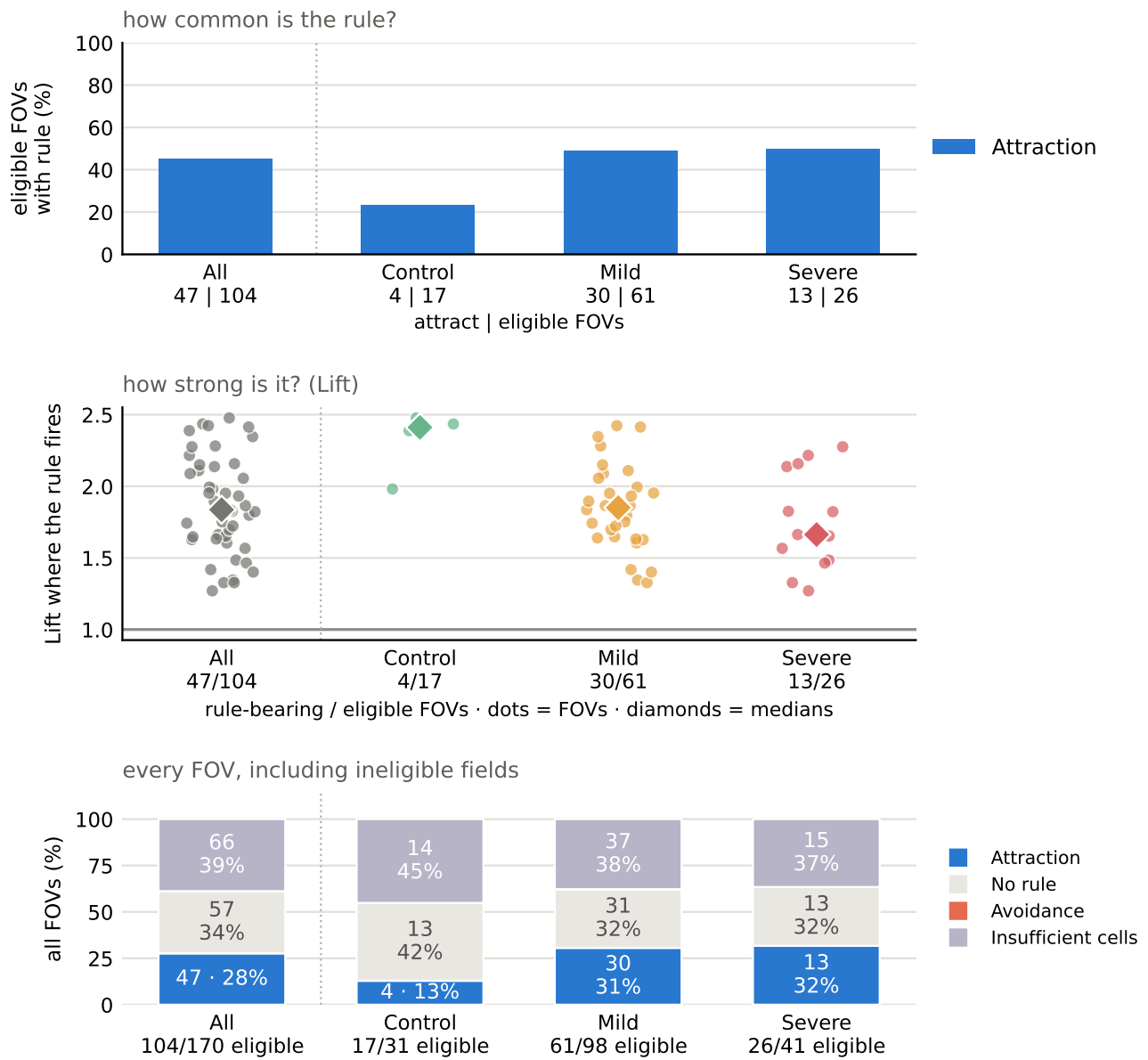


Figure 8: Calculations follow Figure 6. Attraction occurs in 47/104 eligible FOVs and 27/42 eligible patients, versus 30/104 fields and 17/42 patients in reverse. The recurrence across organs and patients makes this useful as a method check. Endothelial-to-mural developmental signaling [8] does not predict the stronger spatial arrow.

The duodenal examples show where this recurring vascular rule is found.

Duodenum SMV → Endothelial — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: each observed state near median Lift; typical eligible no-rule FOV

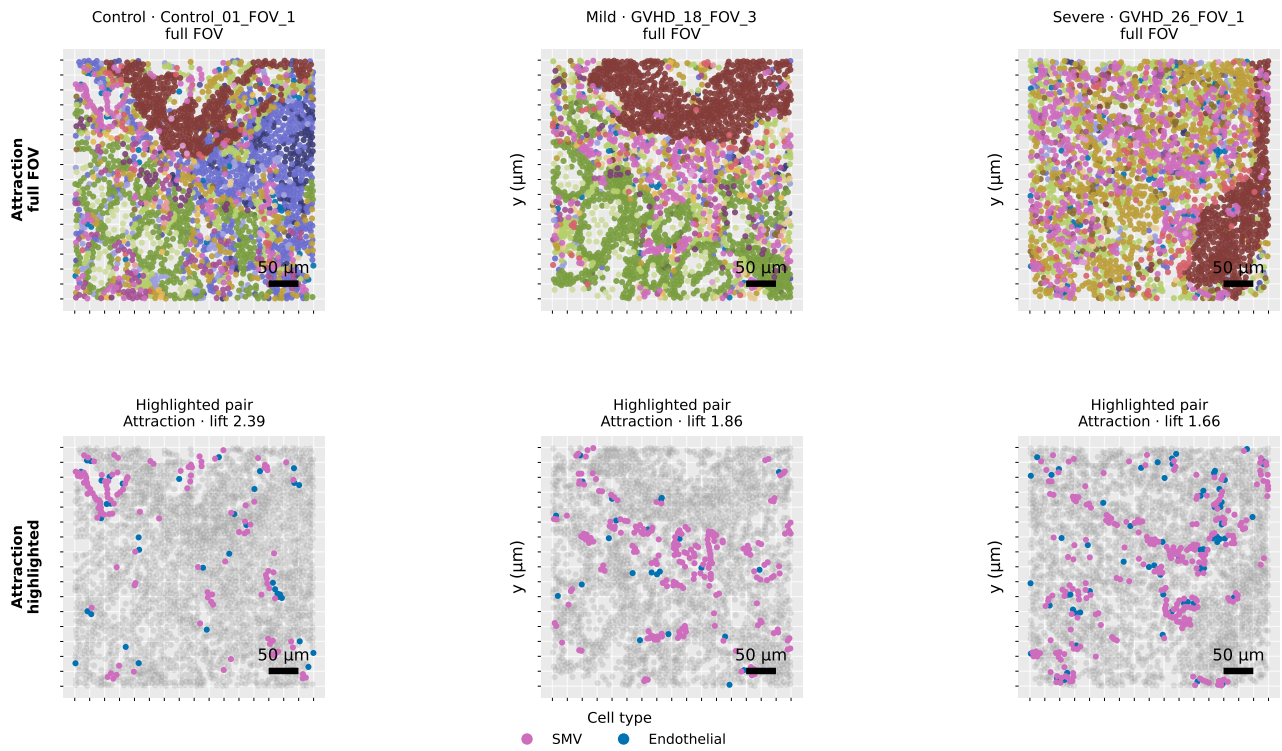


Figure 9: Maps and selection follow Figure 7.

Paneth → Epithelial is common in eligible fields, consistent with a restricted epithelial niche [9].

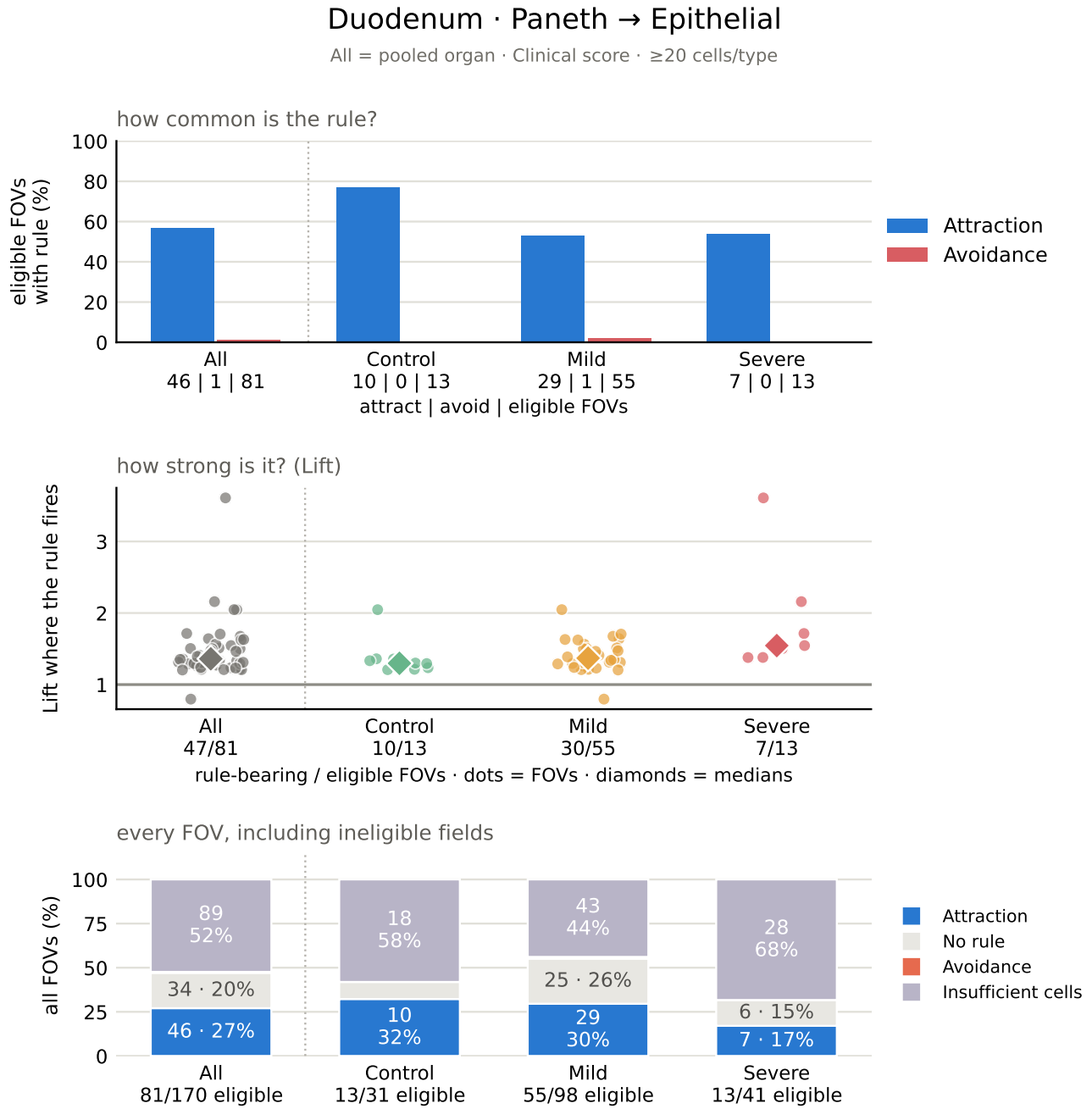


Figure 10: Calculations follow Figure 6. Attraction occurs in 46/81 fields versus 8/81 in reverse, across 28/36 versus 7/36 eligible patients. This recovers an asymmetric spatial rule, not Paneth-to-stem-cell signaling: the broad Epithelial label does not identify stem cells. The separate all-center benchmark in the notebook gives little positive Paneth-centered excess; its null and sampling differ from mining.

The examples show Paneth cells beside epithelial structures where the rule was mined.

Duodenum Paneth → Epithelial — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: each observed state near median Lift; typical eligible no-rule FOV

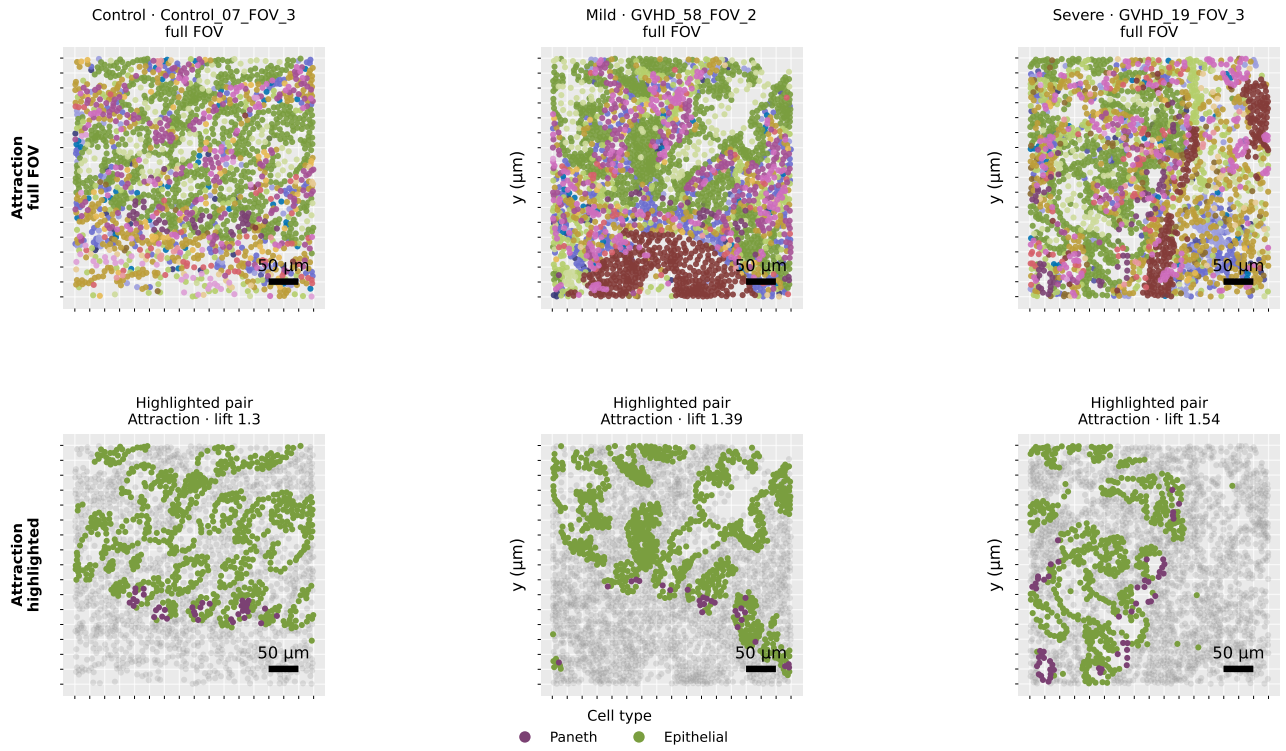


Figure 11: Maps and selection follow Figure 7.

Neuron → Macrophage recurs in colon fields, linking the mined rule to a known neuroimmune partnership [3].

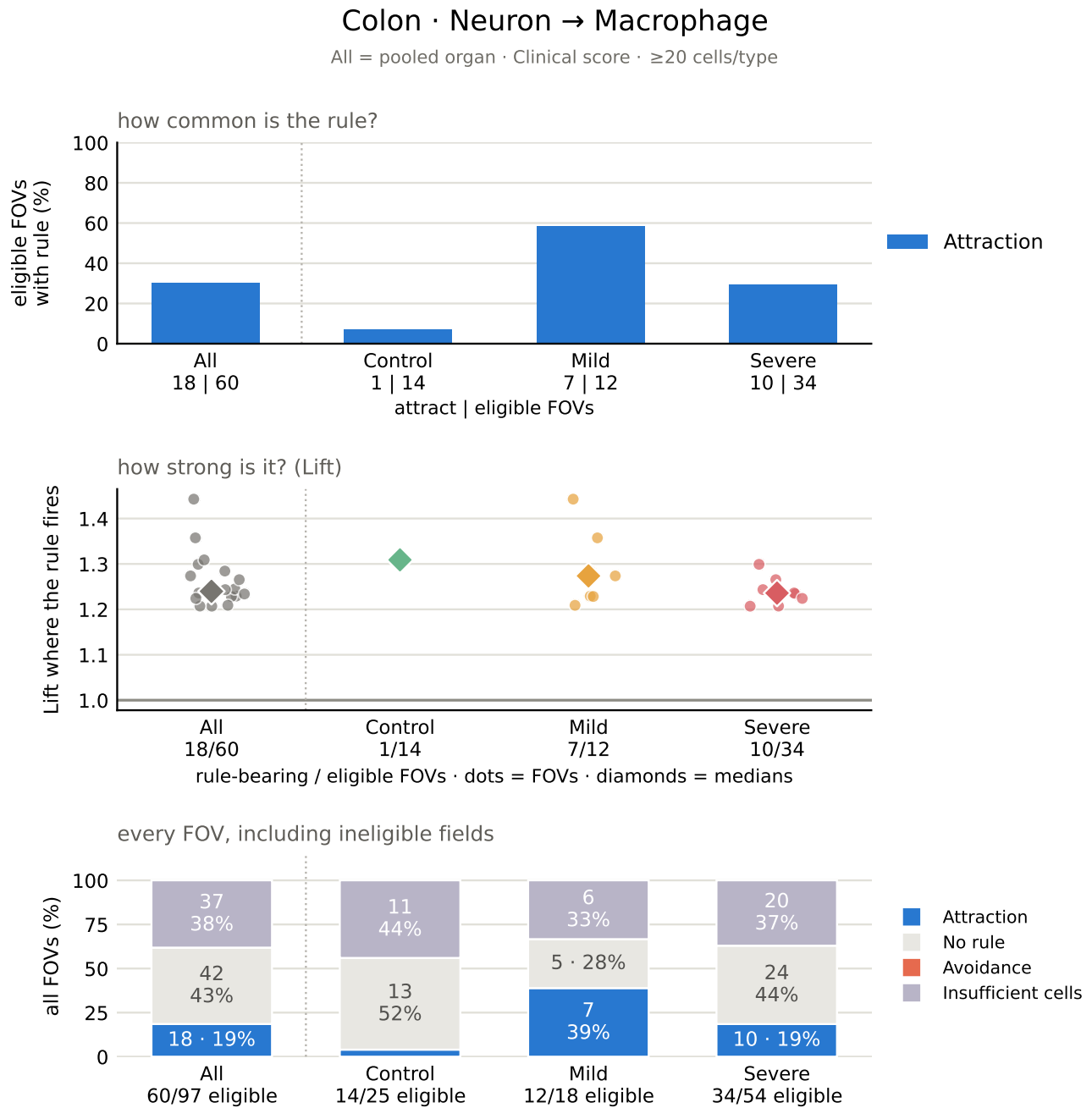


Figure 12: Calculations follow Figure 6. Attraction occurs in 18/60 fields versus 6/60 in reverse. The reference describes reciprocal BMP2/CSF1 signaling; these broad labels and spatial rules cannot identify its molecular direction. The notebook's separate spatial-excess check favors the other arrow, reinforcing that distinction.

The highlighted fields show the macrophage–neuron relationship behind the occurrence counts.

Colon Neuron → Macrophage — representative FOVs

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: each observed state near median Lift; typical eligible no-rule FOV

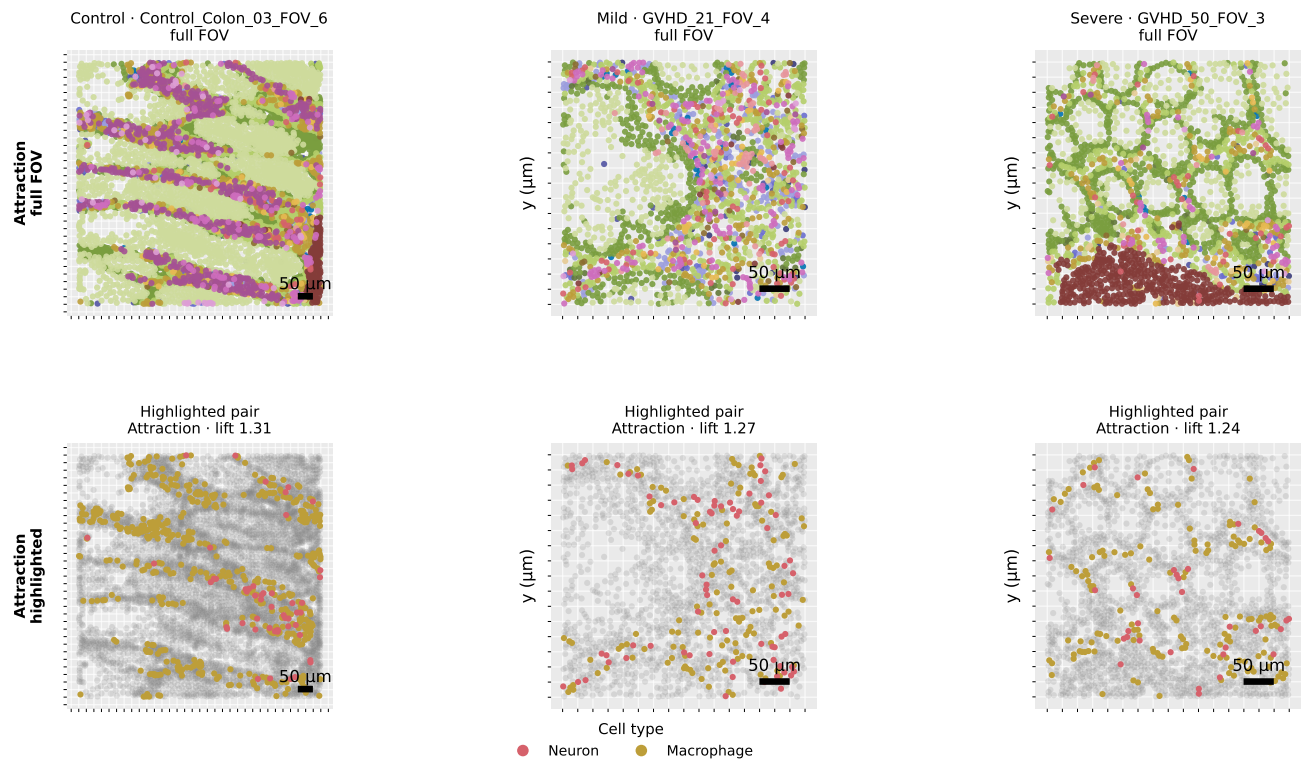


Figure 13: Maps and selection follow Figure 7.

Macrophage → Fibroblast attraction recurs in 19 duodenal patients, giving a localized example of a literature-motivated partnership [4].

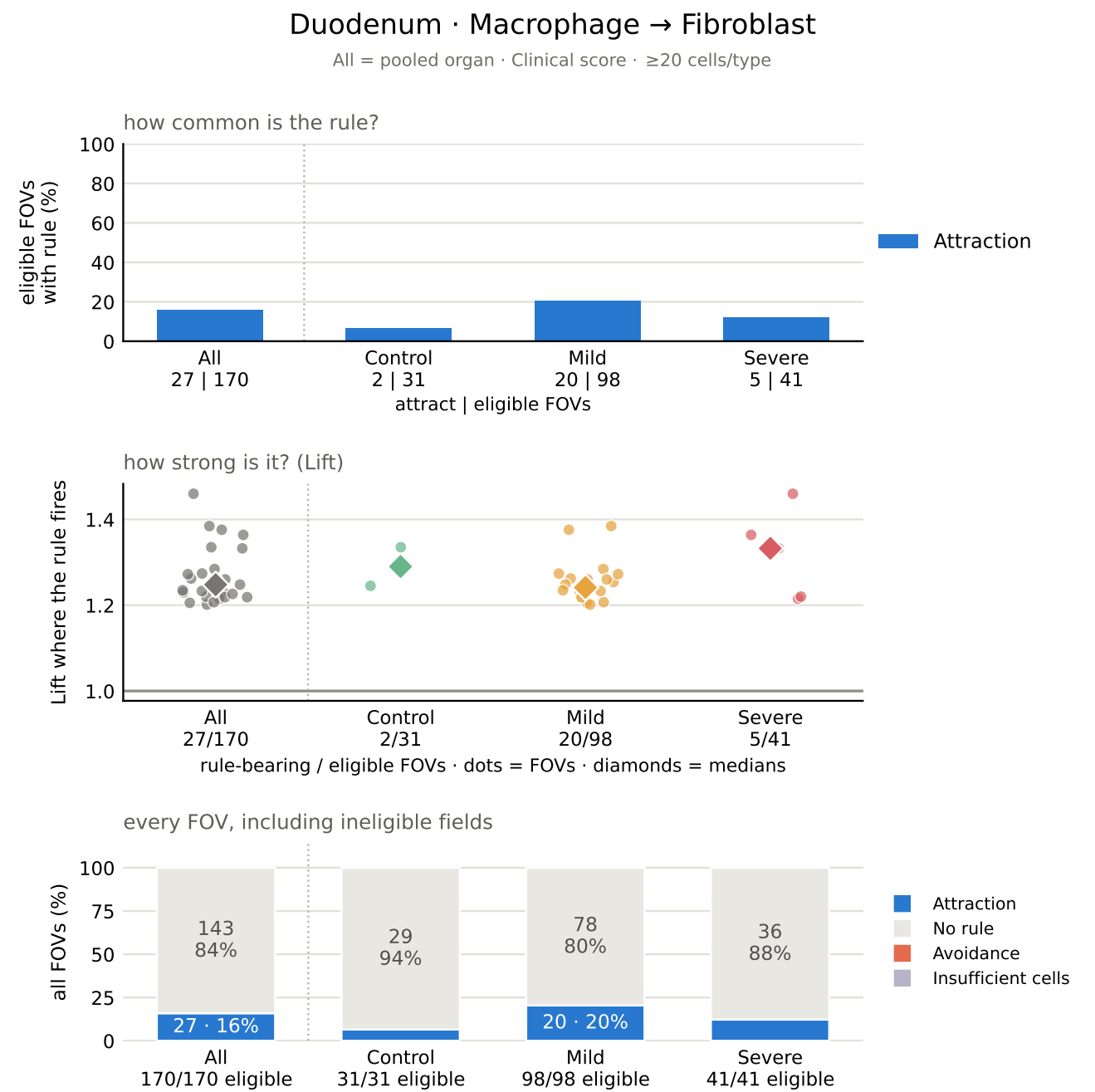


Figure 14: Calculations follow Figure 6. Attraction occurs in 27/170 eligible fields across 19/49 patients, versus 25/170 fields across 20/49 patients in reverse. Lift is modest; this is recurrent attraction in a minority of fields, not a preferred arrow or a universal niche. The reference concerns specific intestinal subsets, whereas these labels are broad. The result does not establish GM-CSF signaling.

The representative fields show where macrophages and fibroblasts form these local arrangements.

Duodenum Macrophage → Fibroblast — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: each observed state near median Lift; typical eligible no-rule FOV

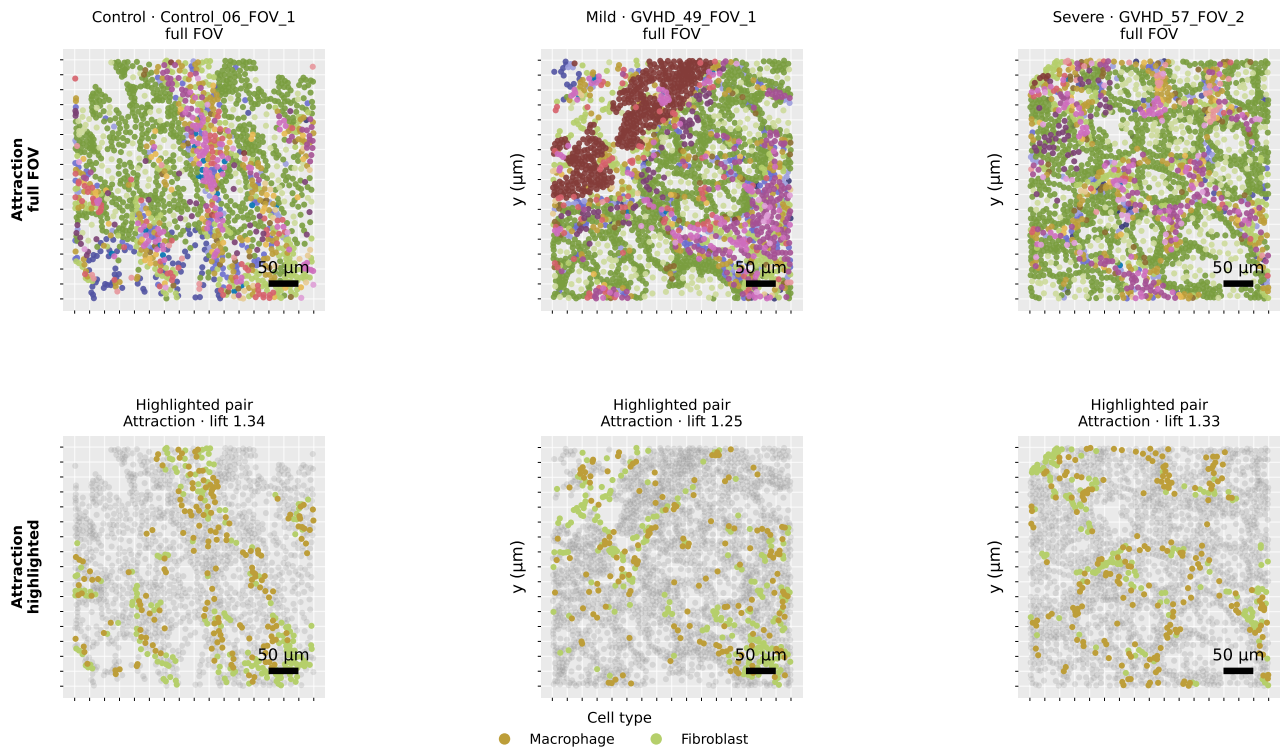


Figure 15: Maps and selection follow Figure 7. Typical attracting fields are shown, not the many eligible fields without a passing rule.

2 Paper alignment

These rules were chosen from biological relationships highlighted by the spatial atlas [1], not because they ranked highly in our data.

Duodenal epithelial → goblet attraction is common in every stage, recovering a stable epithelial relationship.

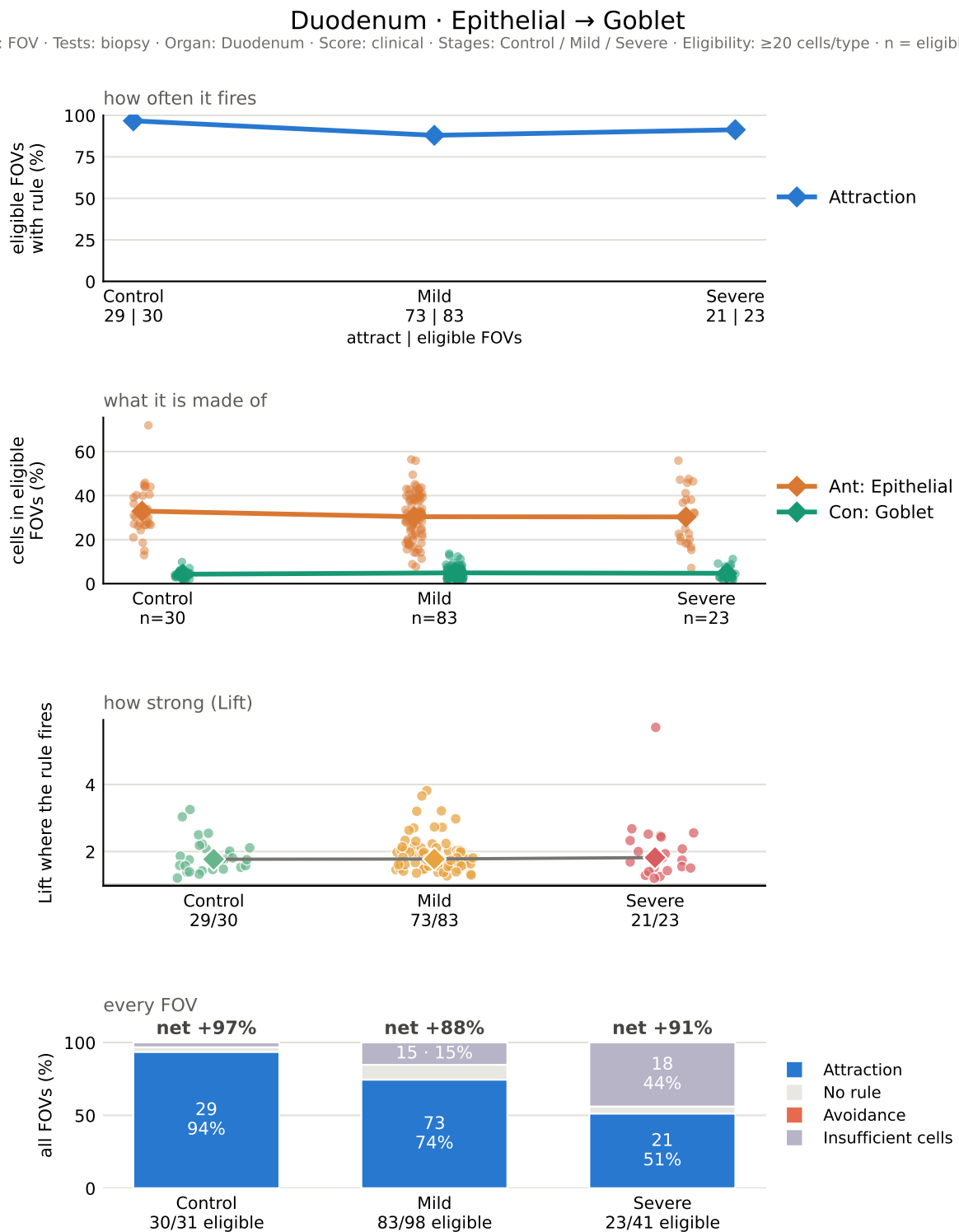


Figure 16: Read as in Figure 3.

The maps show rule-bearing tissue.

Duodenum Epithelial → Goblet — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

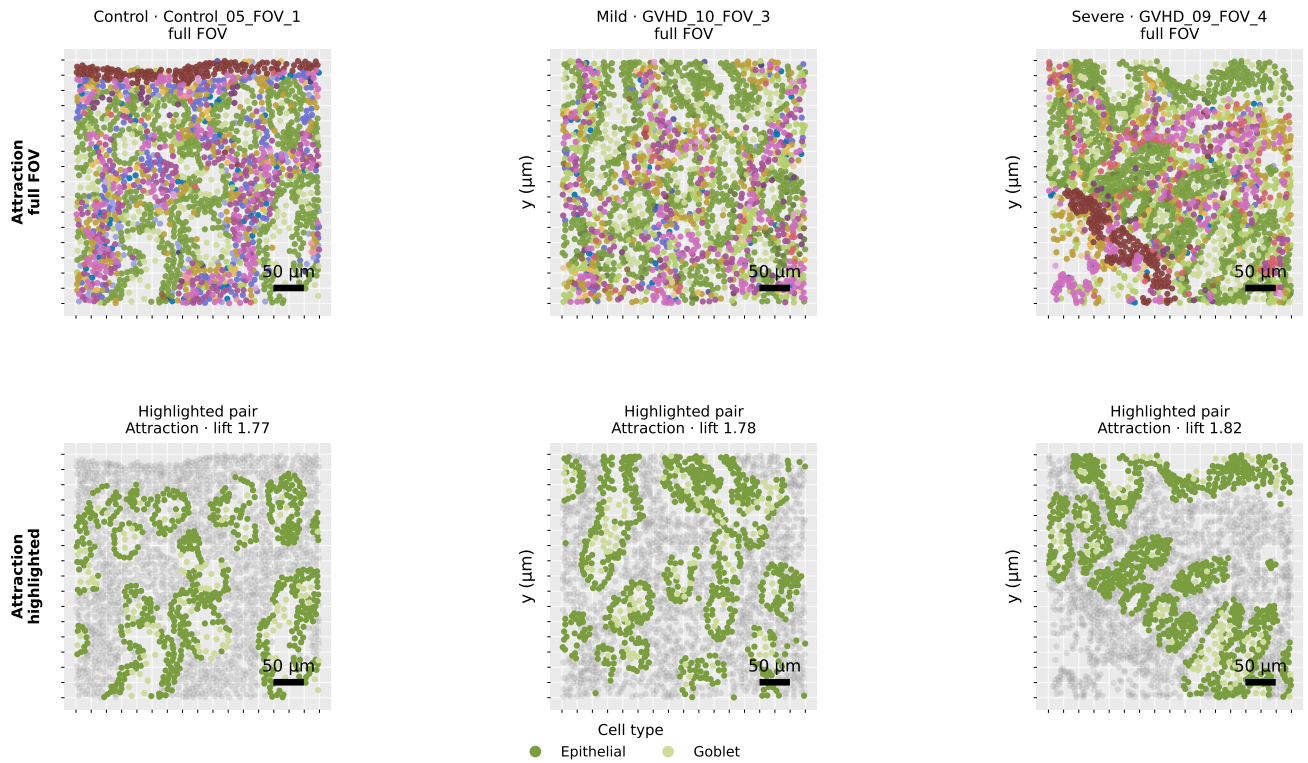


Figure 17: This is the first expanded FOV view. Columns are stages or time windows, one rule state per figure. Each example appears first as a full FOV and then with only the rule's cell types highlighted. Rule-bearing examples are nearest their state's median Lift; a no-rule figure uses eligible comparison fields.

Eligible fields where this relationship was not mined give the contrast.

Duodenum Epithelial → Goblet — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

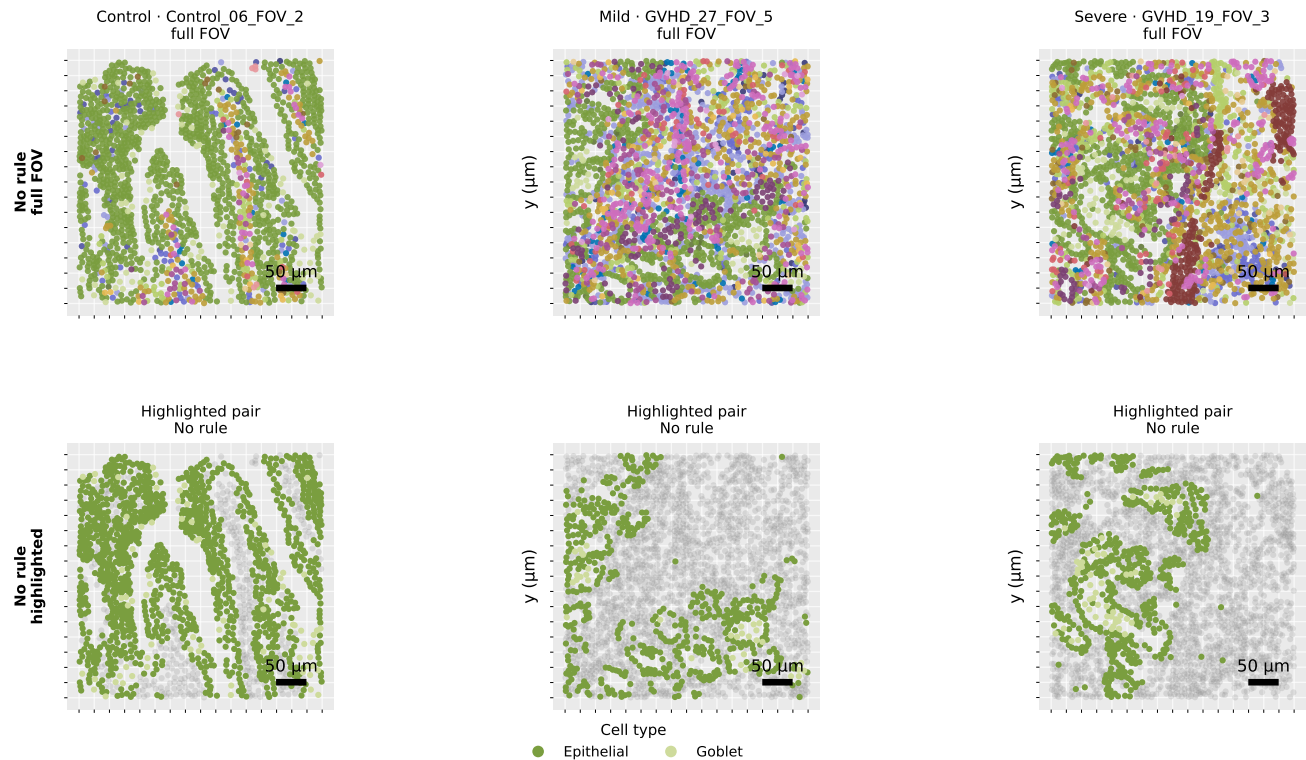


Figure 18: Maps and example selection follow Figure 17.

Paneth → Epithelial becomes less frequent but remains attractive, matching its restricted crypt niche [9].

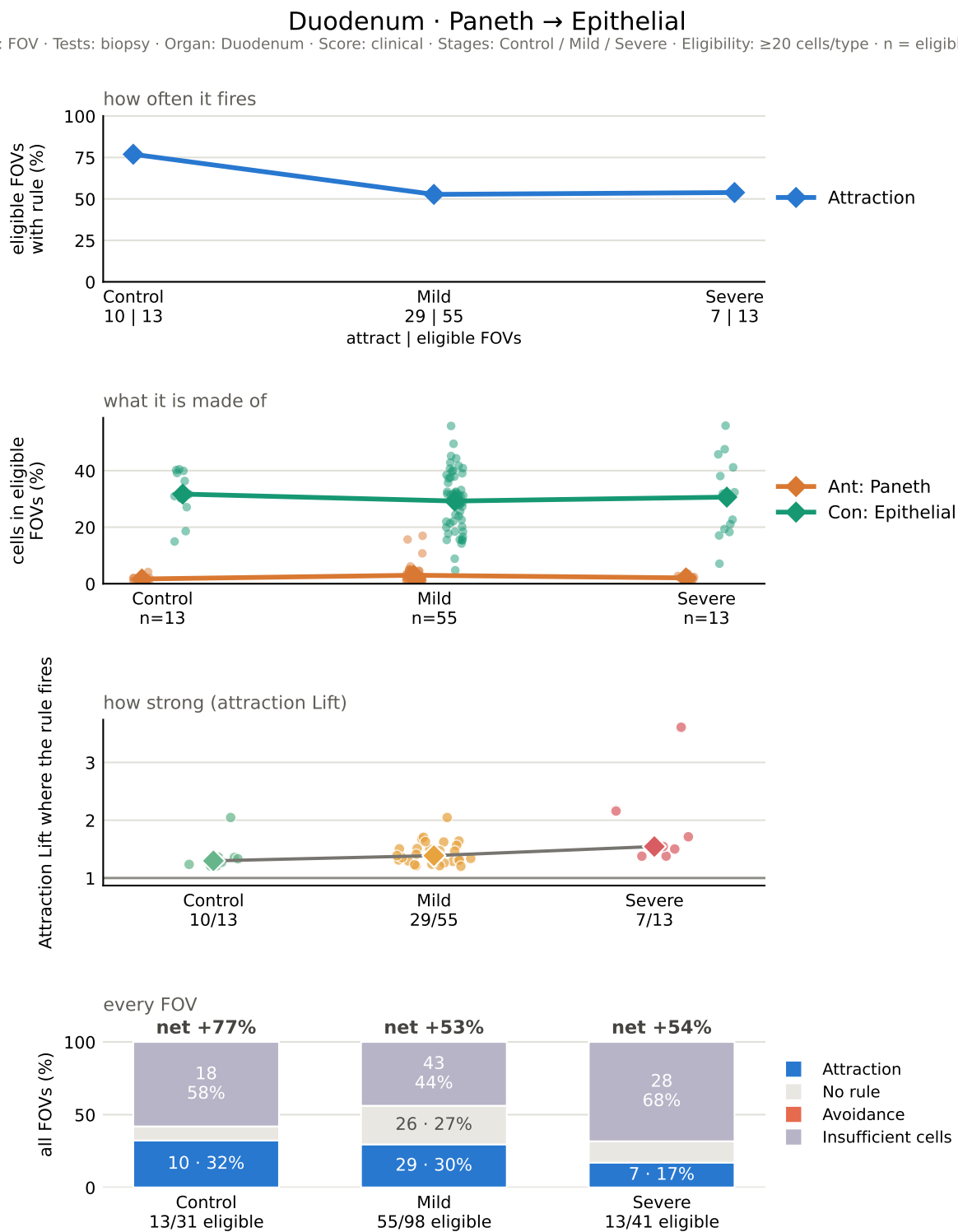


Figure 19: Read as in Figure 3. The strength row contains attraction occurrences only.

The examples place Paneth cells beside epithelial structures, making the mined relationship visible.

Duodenum Paneth → Epithelial — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

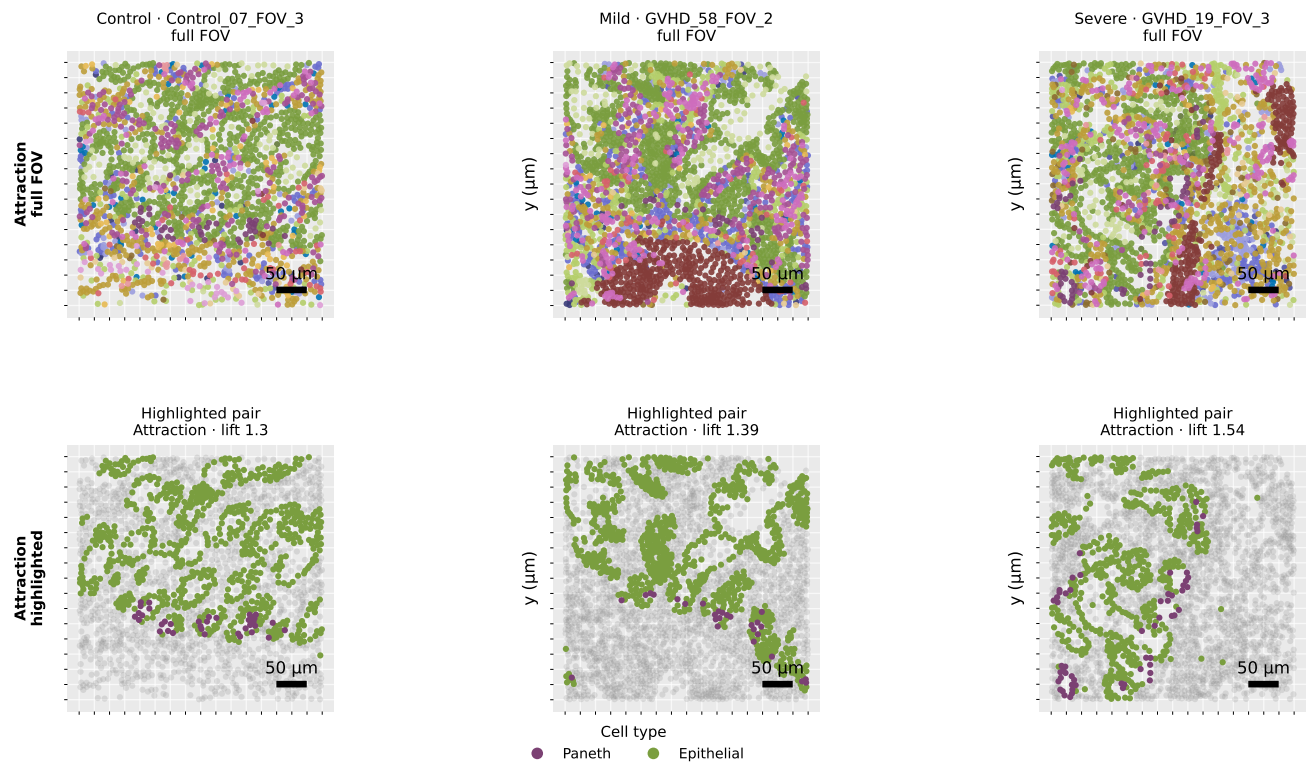


Figure 20: Maps and example selection follow Figure 17.

Goblet → CD8T avoidance rises with severity, revealing a spatial change in the atlas-motivated epithelial-immune pair.

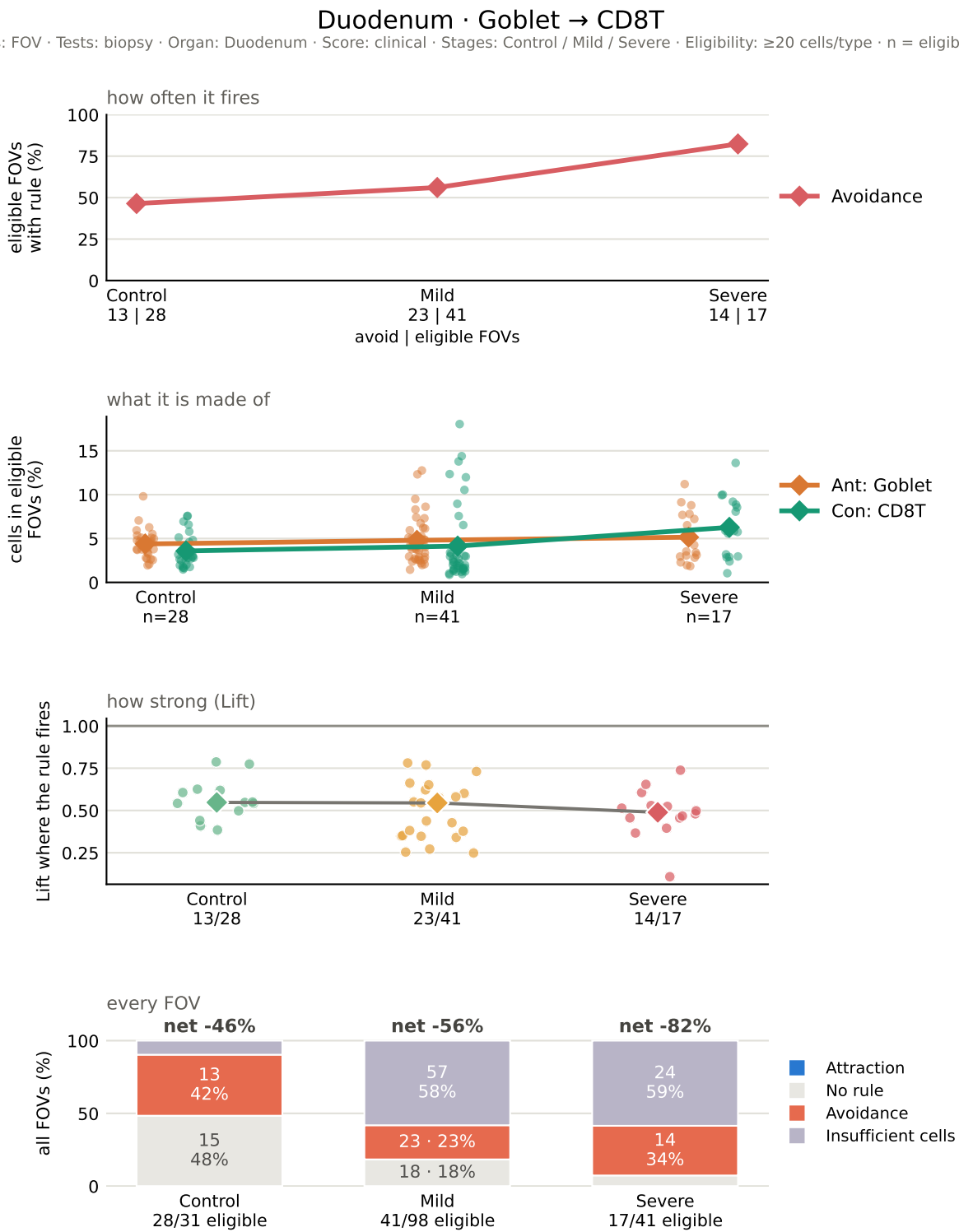


Figure 21: Read as in Figure 3.

The representative fields make the changing epithelial-immune arrangement visible.

Duodenum Goblet → CD8T — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

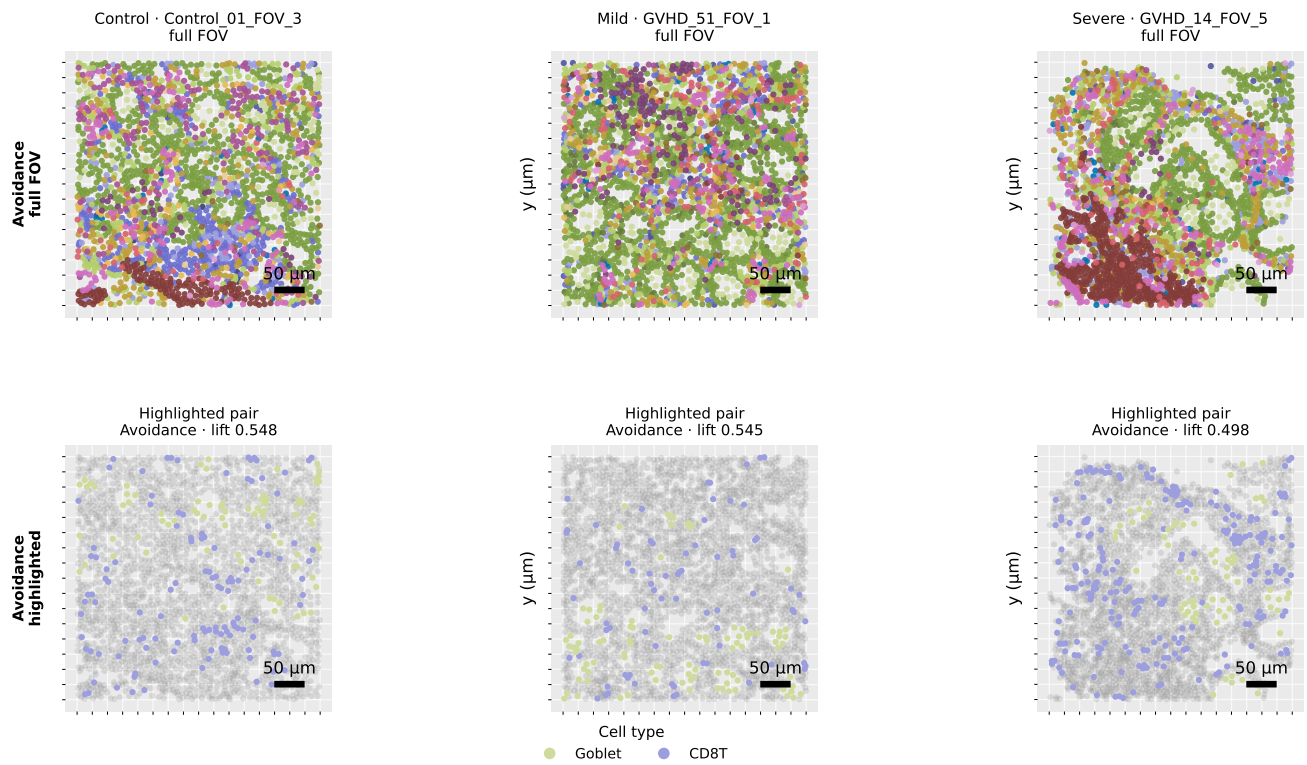


Figure 22: Maps and example selection follow Figure 17.

3 New findings

These candidates were selected automatically by the largest Control–Severe prevalence gaps, with attraction and avoidance ranked separately. Mild-only peaks are excluded.

3.1 Colonic immune relationships weaken with severity

CD4T → Plasma attraction falls from 14/25 Control fields to 1/24 Severe fields.

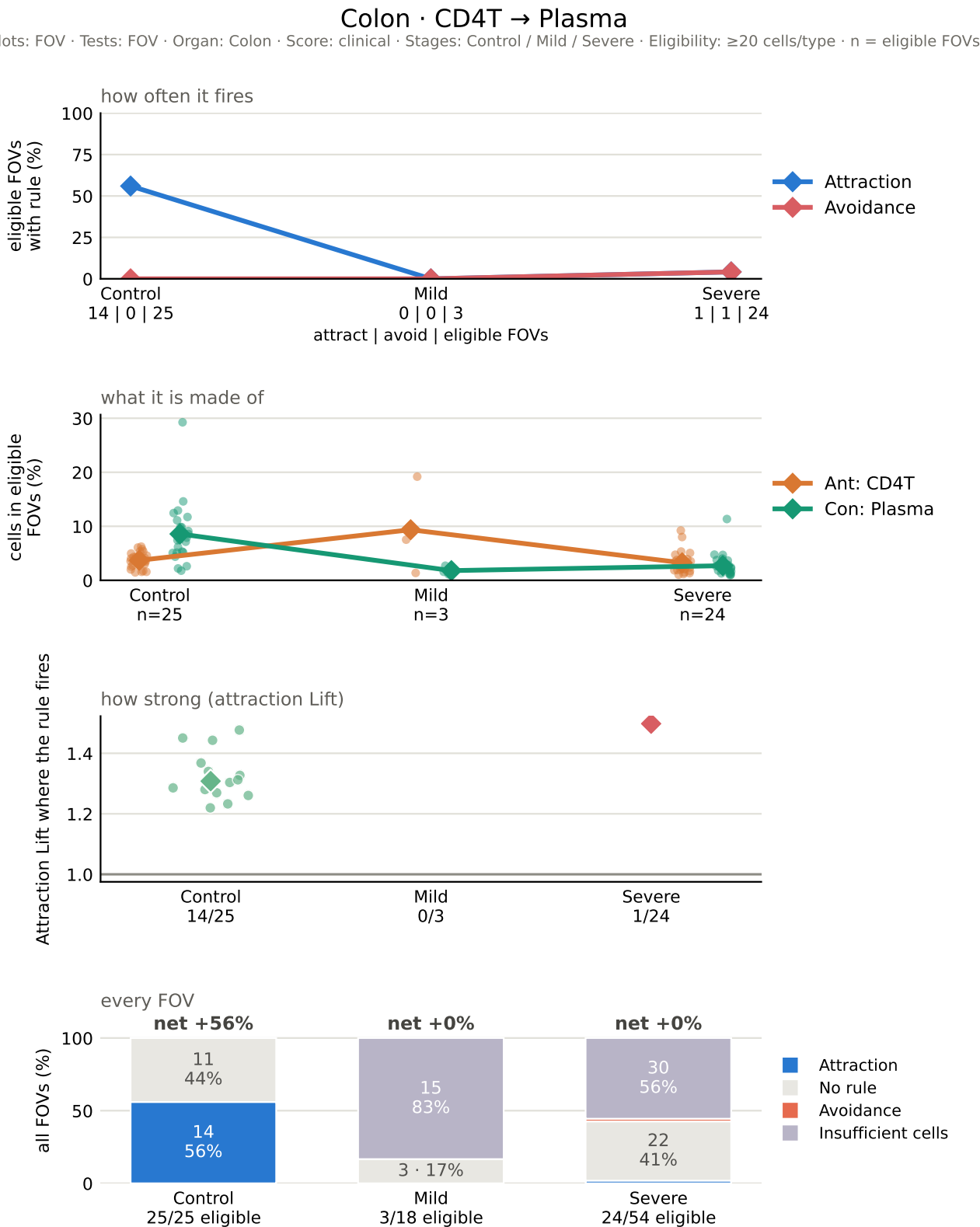


Figure 23: Read as in Figure 3. Selection uses the 52-point Control–Severe attraction gap.

The tissue examples show the attraction.

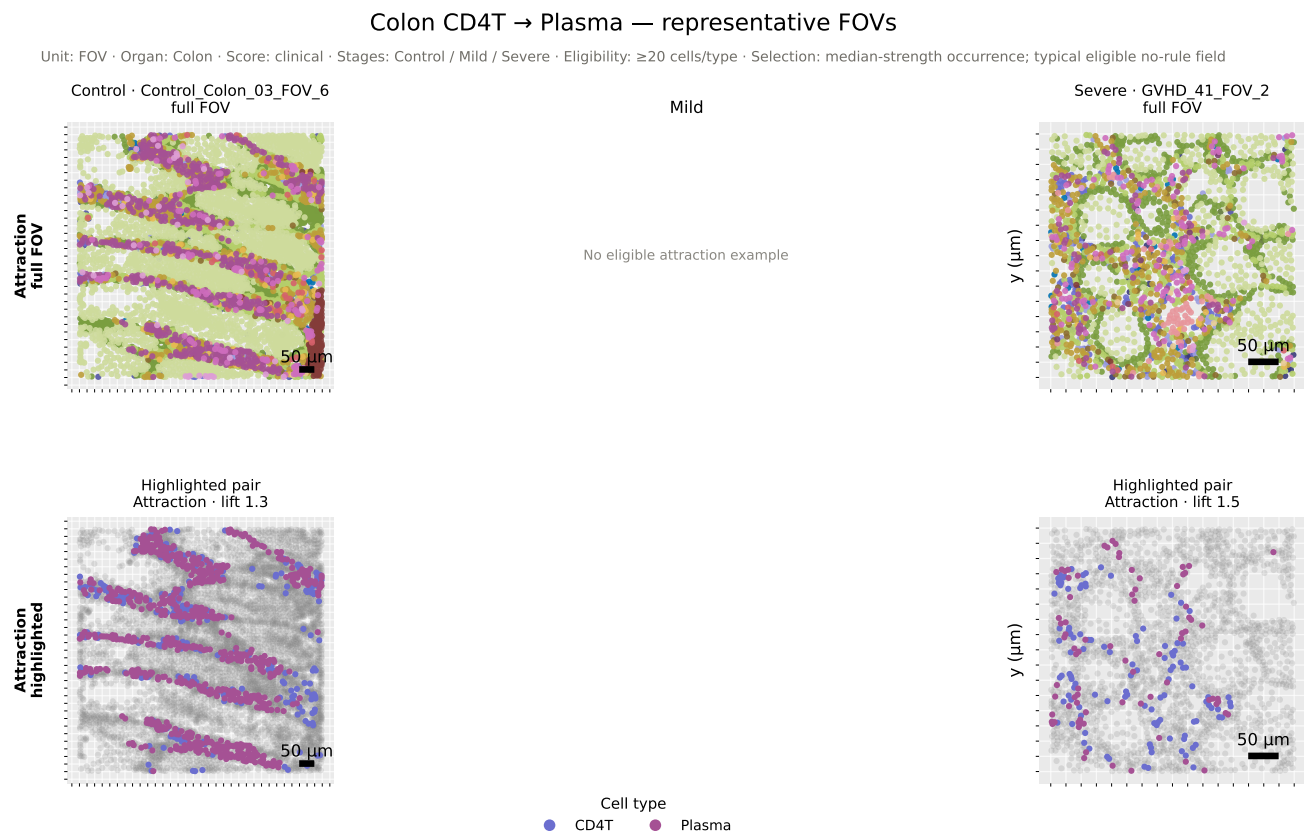


Figure 24: Maps and example selection follow Figure 17.

Eligible fields where the rule is absent give the contrast.

Colon CD4T → Plasma — representative FOVs

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

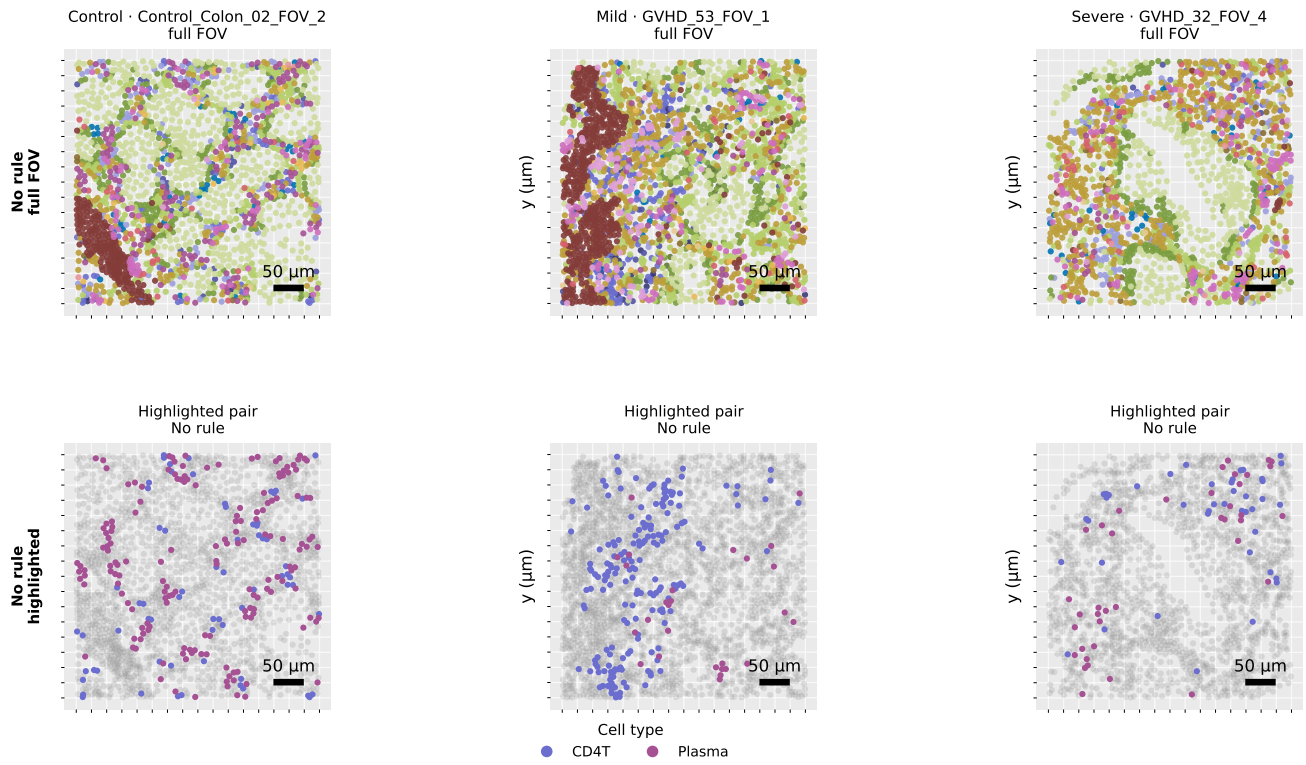


Figure 25: Maps and example selection follow Figure 17.

Goblet → ImmuneOther avoidance falls from 10/11 to 7/16 eligible fields, although the immune label is broad.

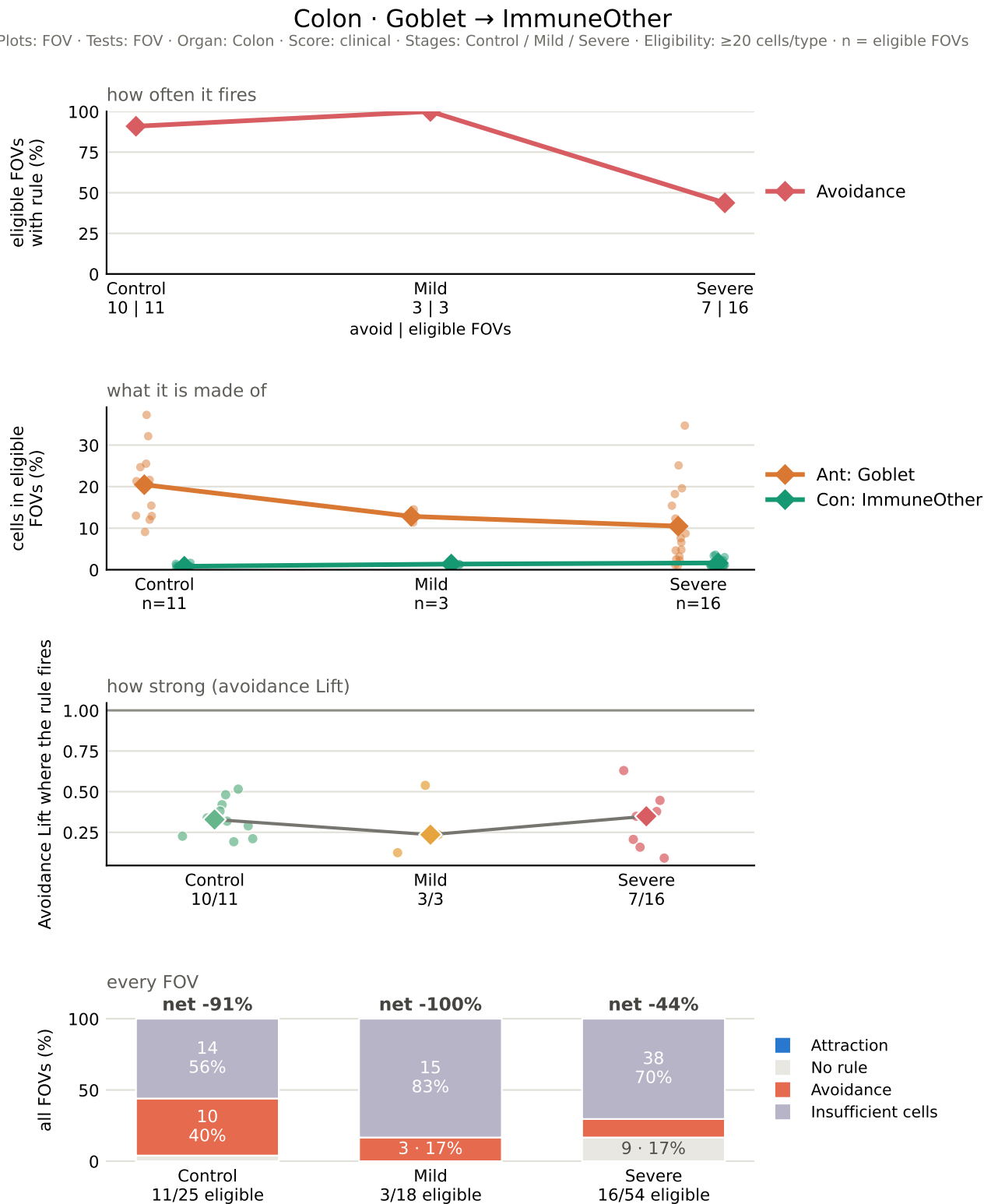


Figure 26: Read as in Figure 3. Selection uses the 47-point Control–Severe avoidance gap.

The maps show the epithelial and immune compartments behind this reduced separation.

Colon Goblet → ImmuneOther — representative FOVs

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

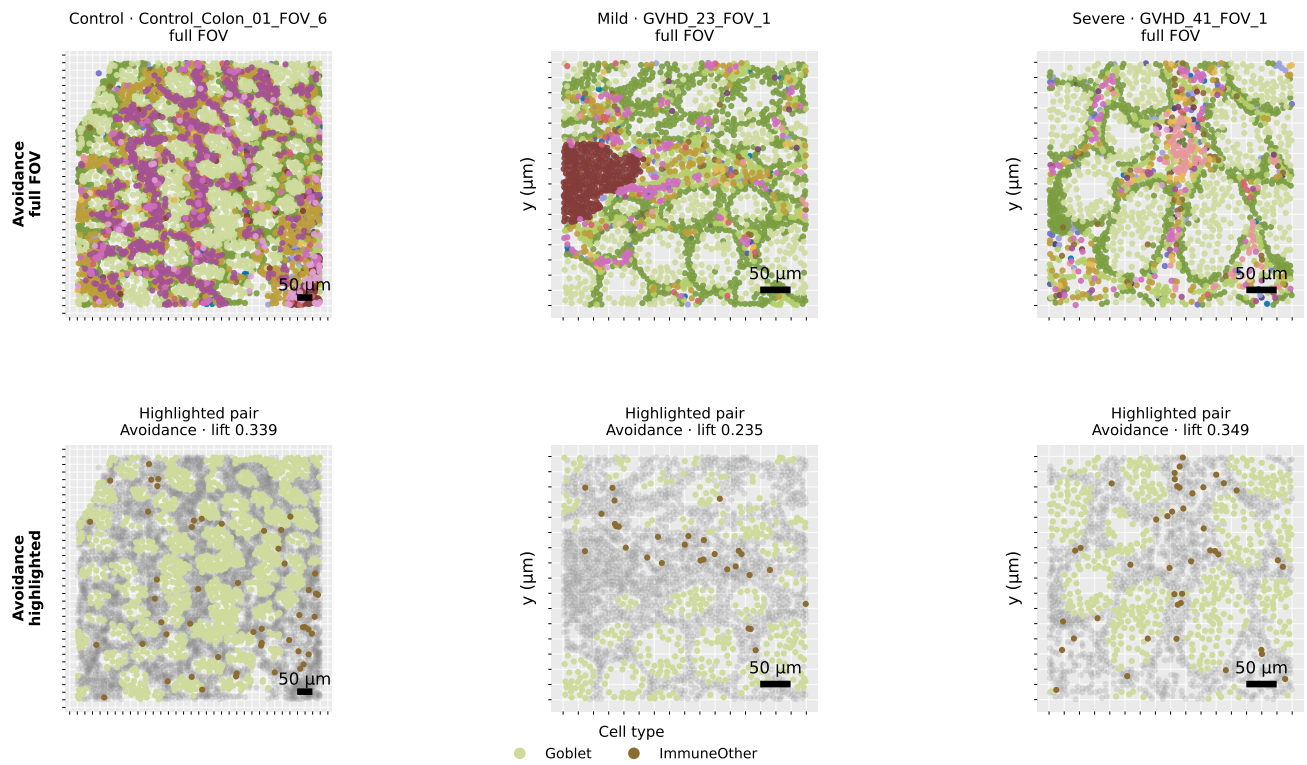


Figure 27: Maps and example selection follow Figure 17.

3.2 Immune cells move relative to the duodenal wall

CD4T → Goblet avoidance falls from 23/30 Control fields to 4/21 Severe fields, suggesting less separation between cells with known inflammatory interactions [5, 6].

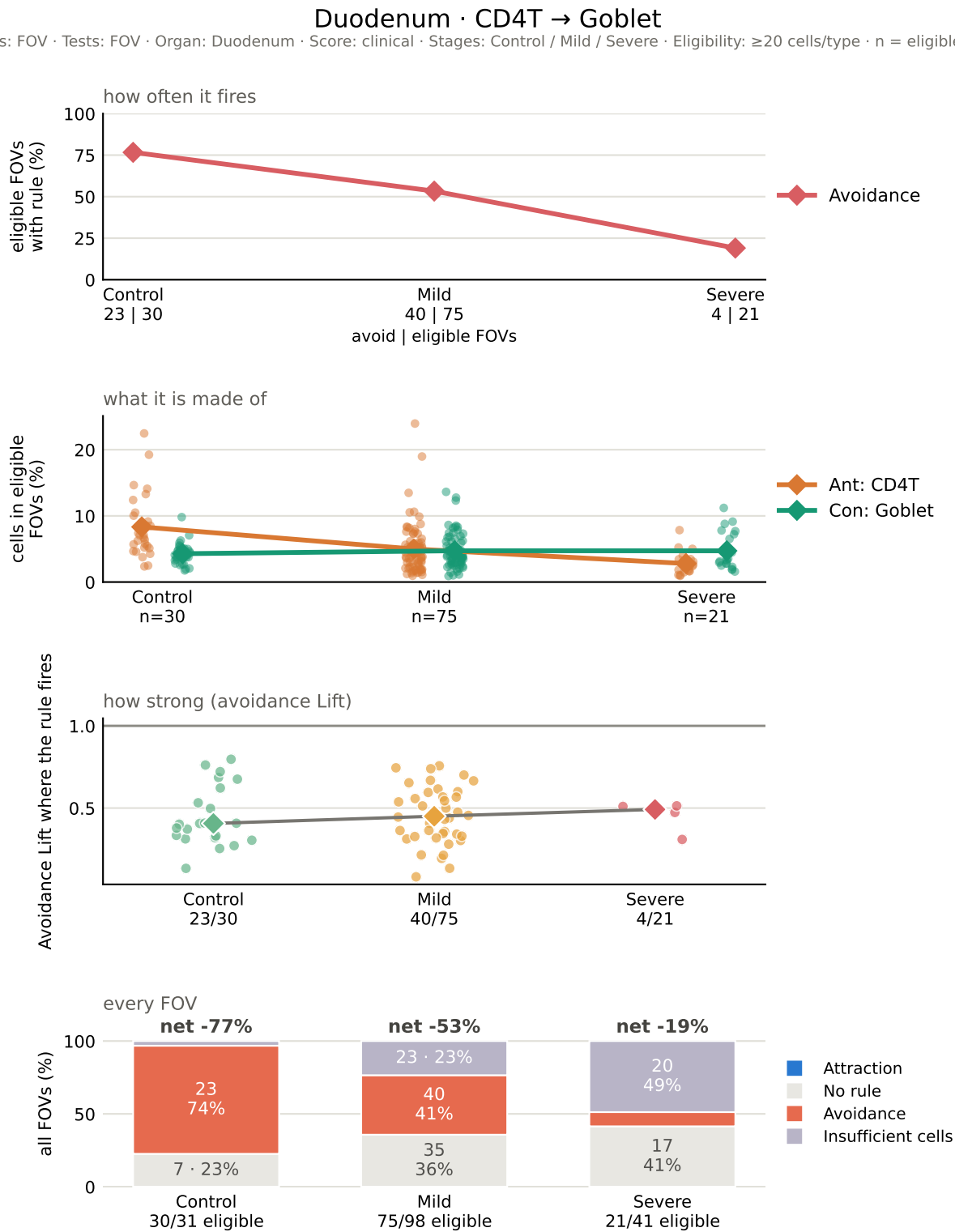


Figure 28: Read as in Figure 3. Selection uses the 58-point Control–Severe avoidance gap.

The maps show the CD4T and goblet-cell positions underlying this change.

Duodenum CD4T → Goblet — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

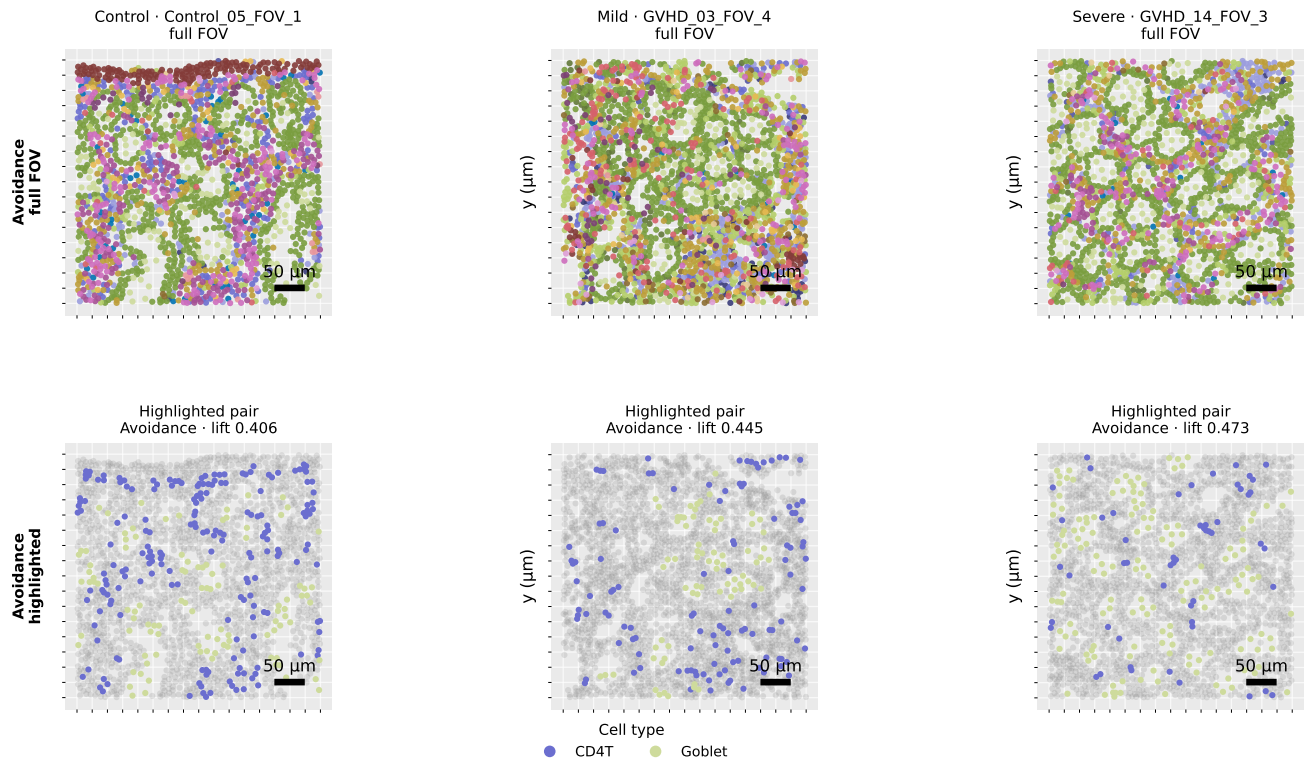


Figure 29: Maps and example selection follow Figure 17.

Muscle → CD4T avoidance rises from 4/18 to 17/25 eligible fields, showing increasing immune–wall separation.

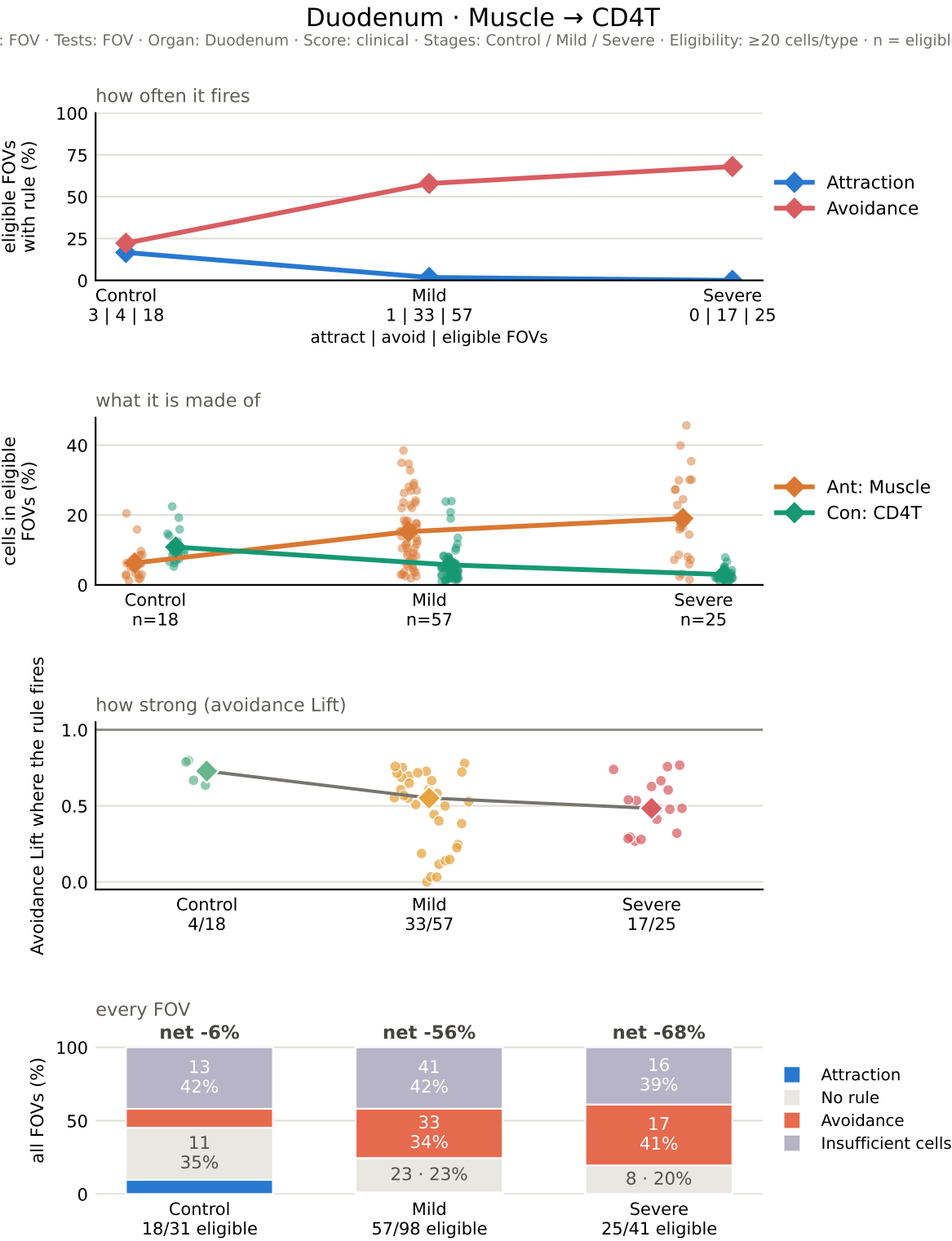


Figure 30: Read as in Figure 3. Selection uses the 46-point Control–Severe avoidance gap.

The representative fields show the growing avoidance.

Duodenum Muscle → CD4T — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

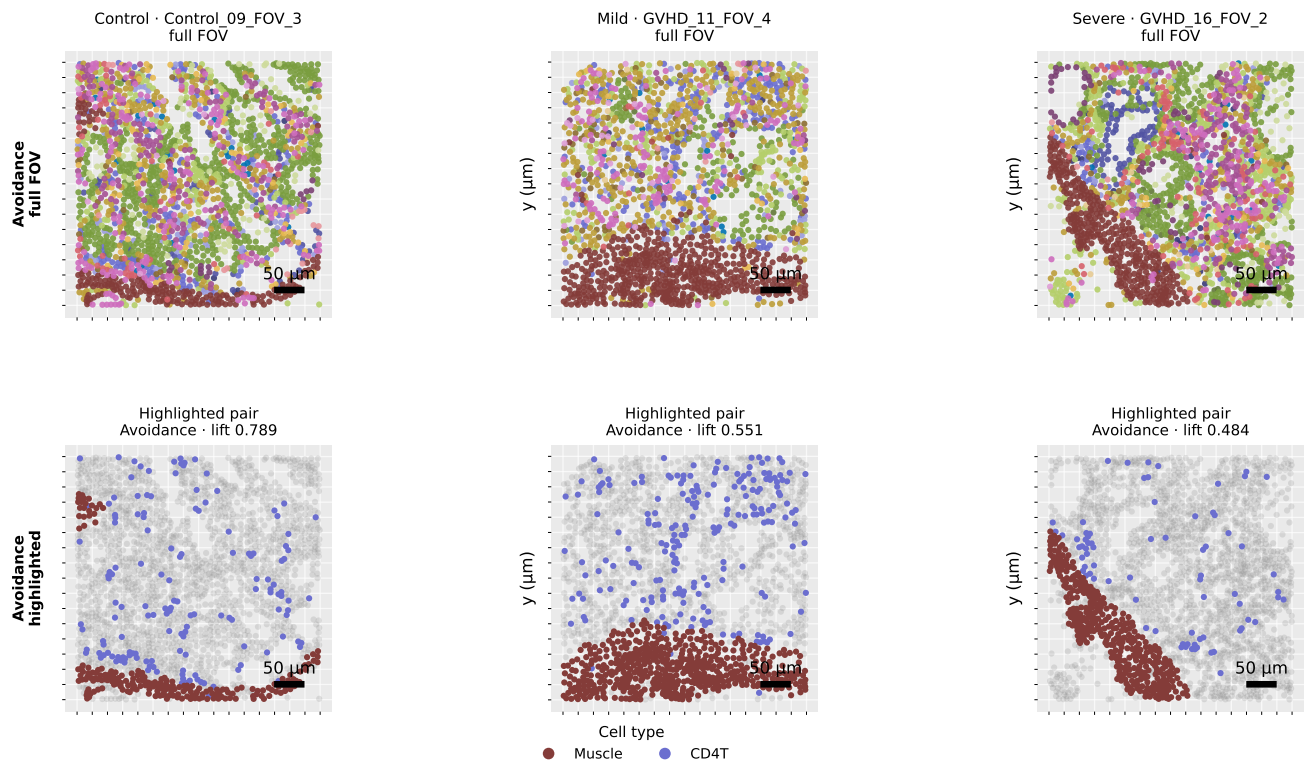


Figure 31: Maps and example selection follow Figure 17.

The less common attraction state is shown separately.

Duodenum Muscle → CD4T — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

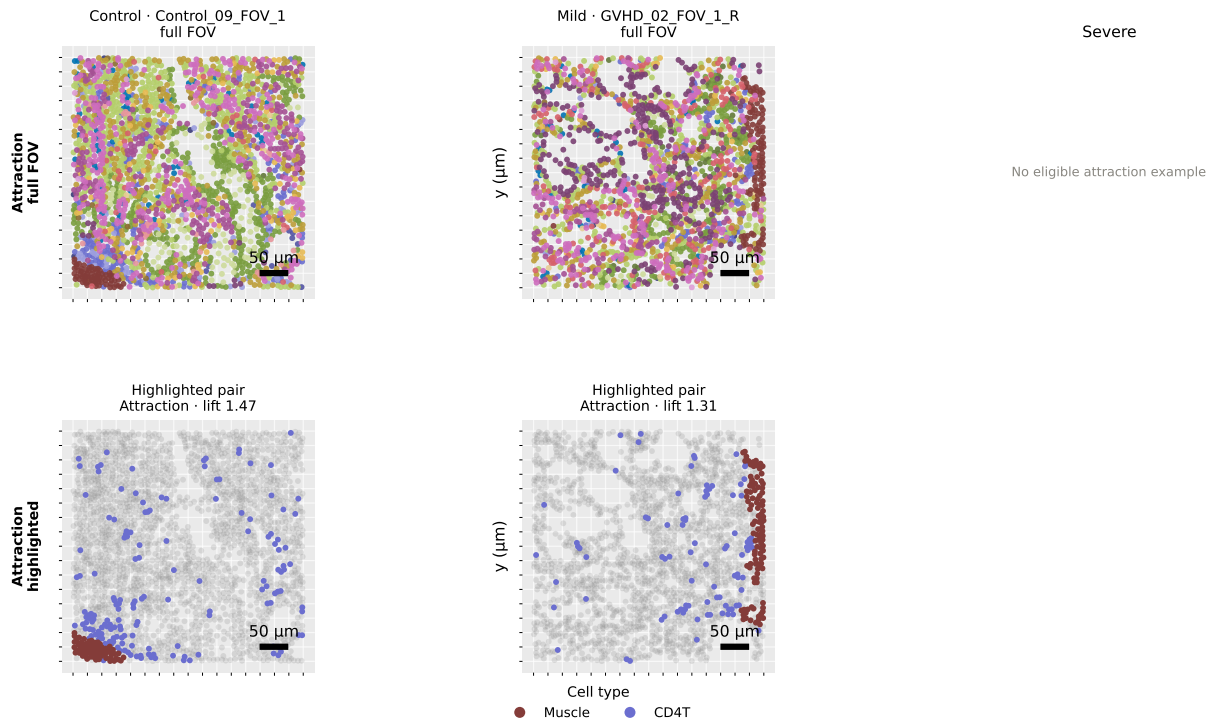


Figure 32: Maps and example selection follow Figure 17.

Eligible no-rule fields complete the comparison.

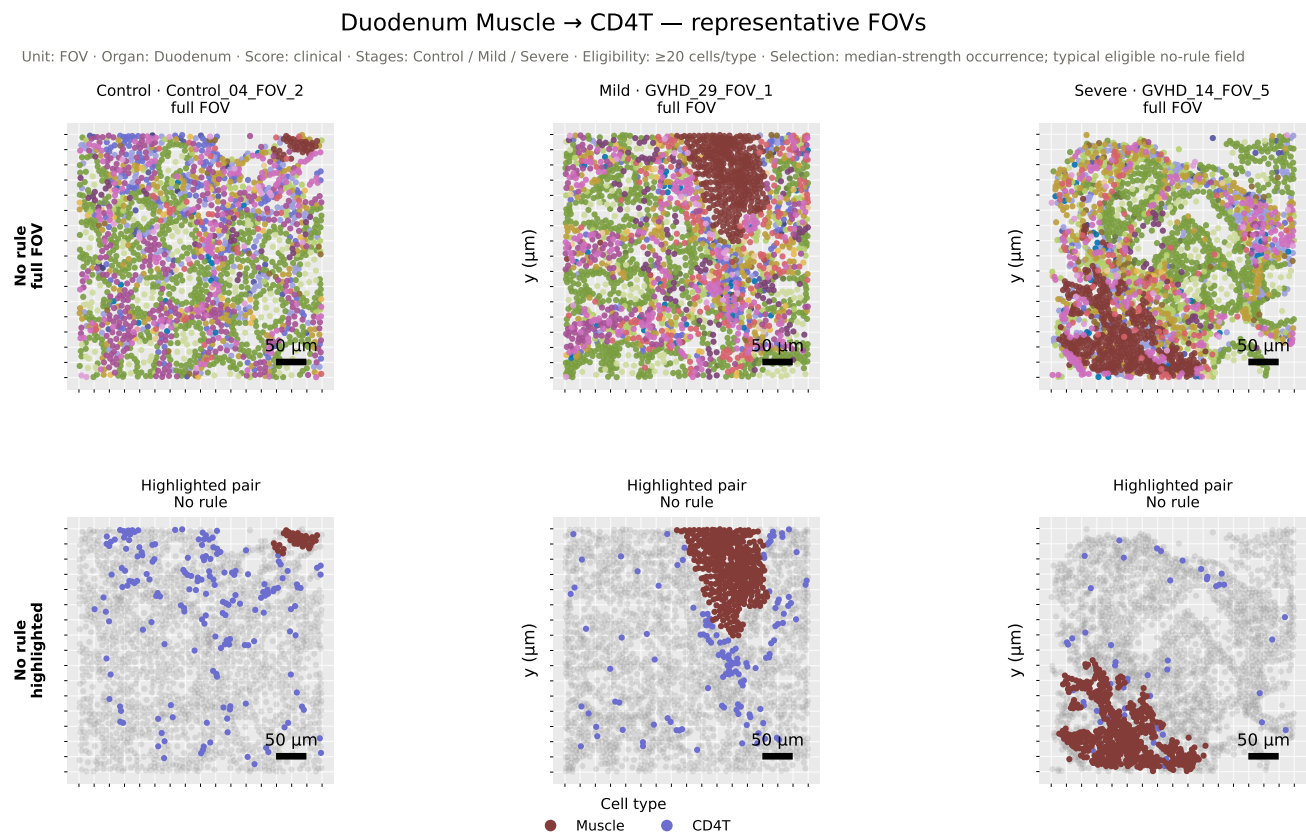


Figure 33: Maps and example selection follow Figure 17.

3.3 A stromal and muscular rearrangement

Fibroblast → Muscle avoidance rises from 6% to 22% to 61%, suggesting progressive stromal separation.

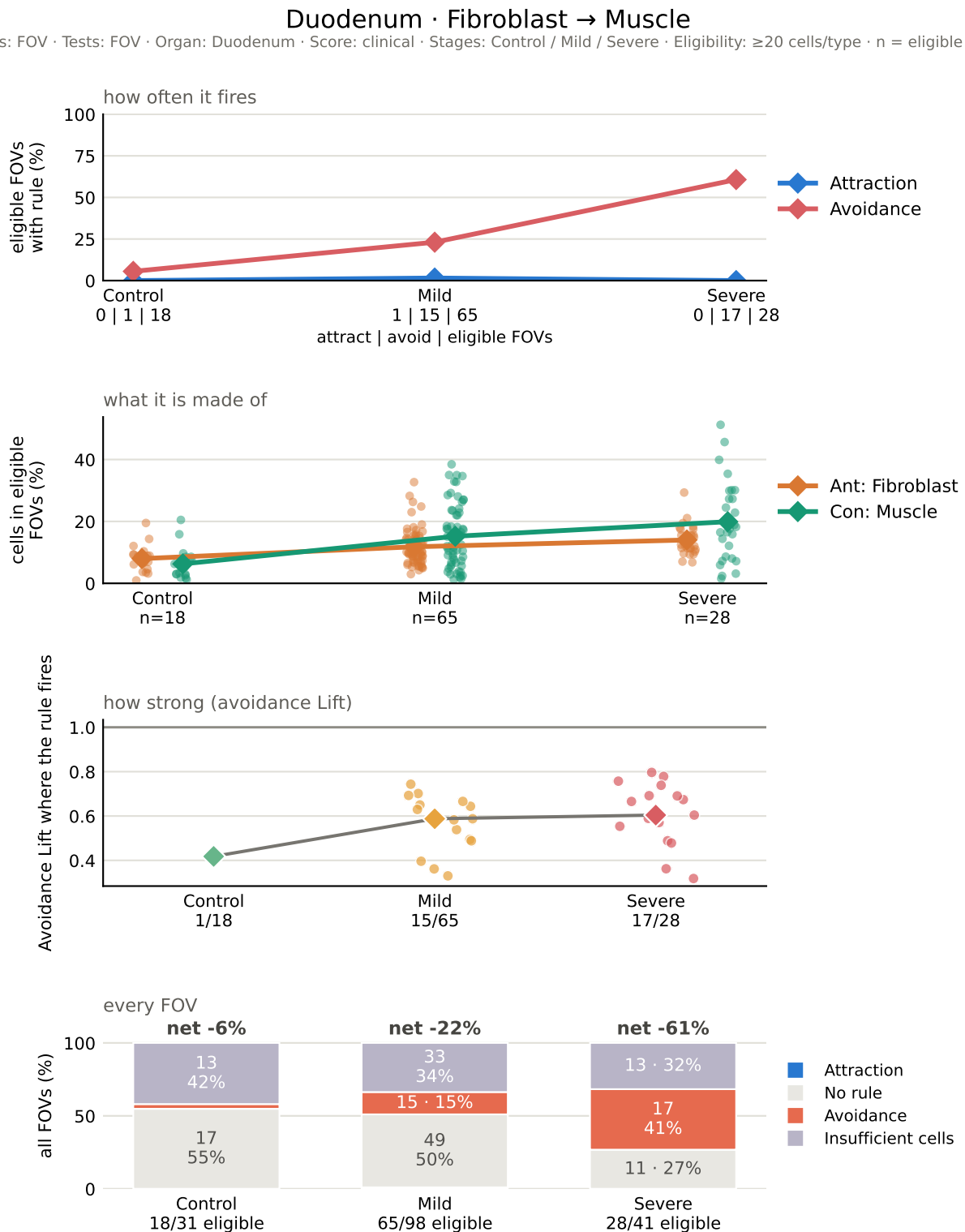


Figure 34: Read as in Figure 3. Selection uses the 55-point Control–Severe avoidance gap. Human intestinal GVHD has documented myofibroblast engraftment [7]; this provides context, not confirmation of this spatial trend.

The examples show the fibroblast and muscle compartments underlying the increasing avoidance.

Duodenum Fibroblast → Muscle — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength occurrence; typical eligible no-rule field

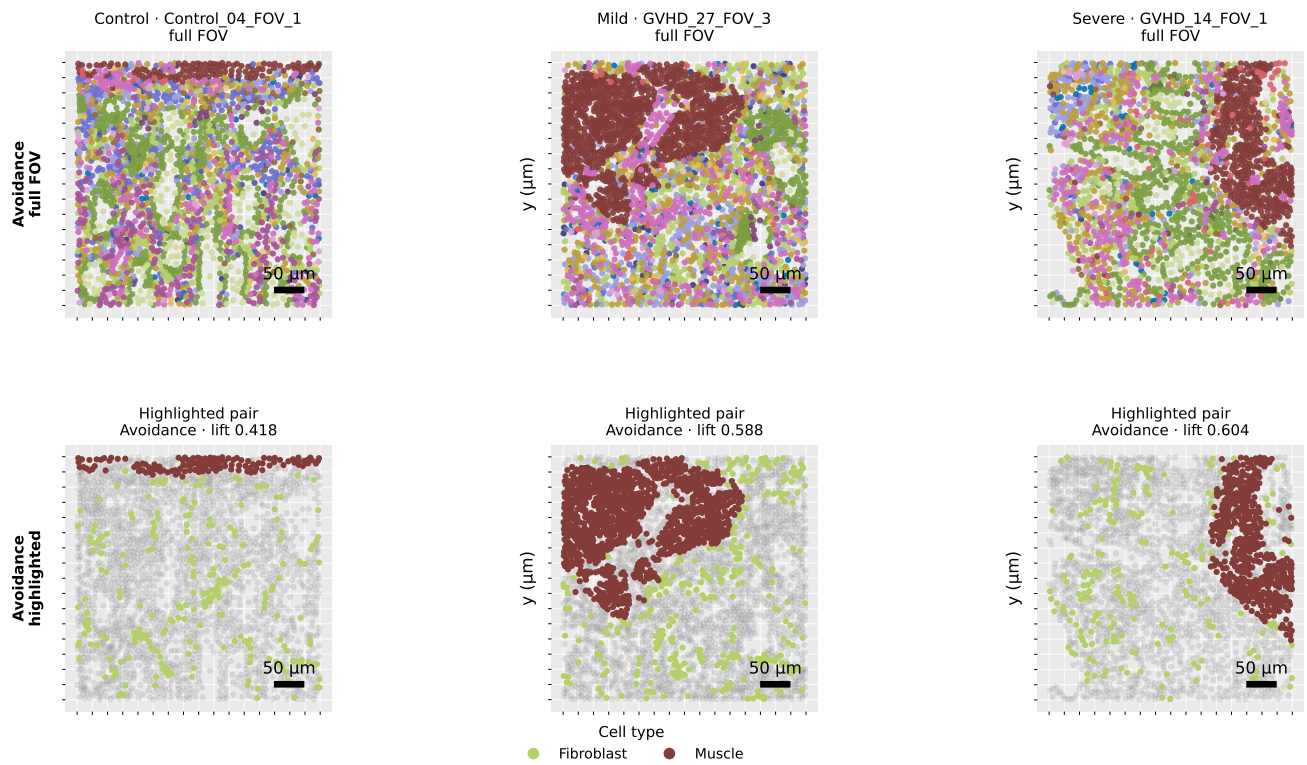


Figure 35: Maps and example selection follow Figure 17.

3.4 Pair rules also change over time after transplantation

The atlas motivates the time windows through immune-cell kinetics, while this analysis asks a different question: whether pair-rule prevalence changes across those windows [1].

Colon Macrophage → CD4T attraction falls from 7/16 fields before day 30 to 0/11 after day 100.

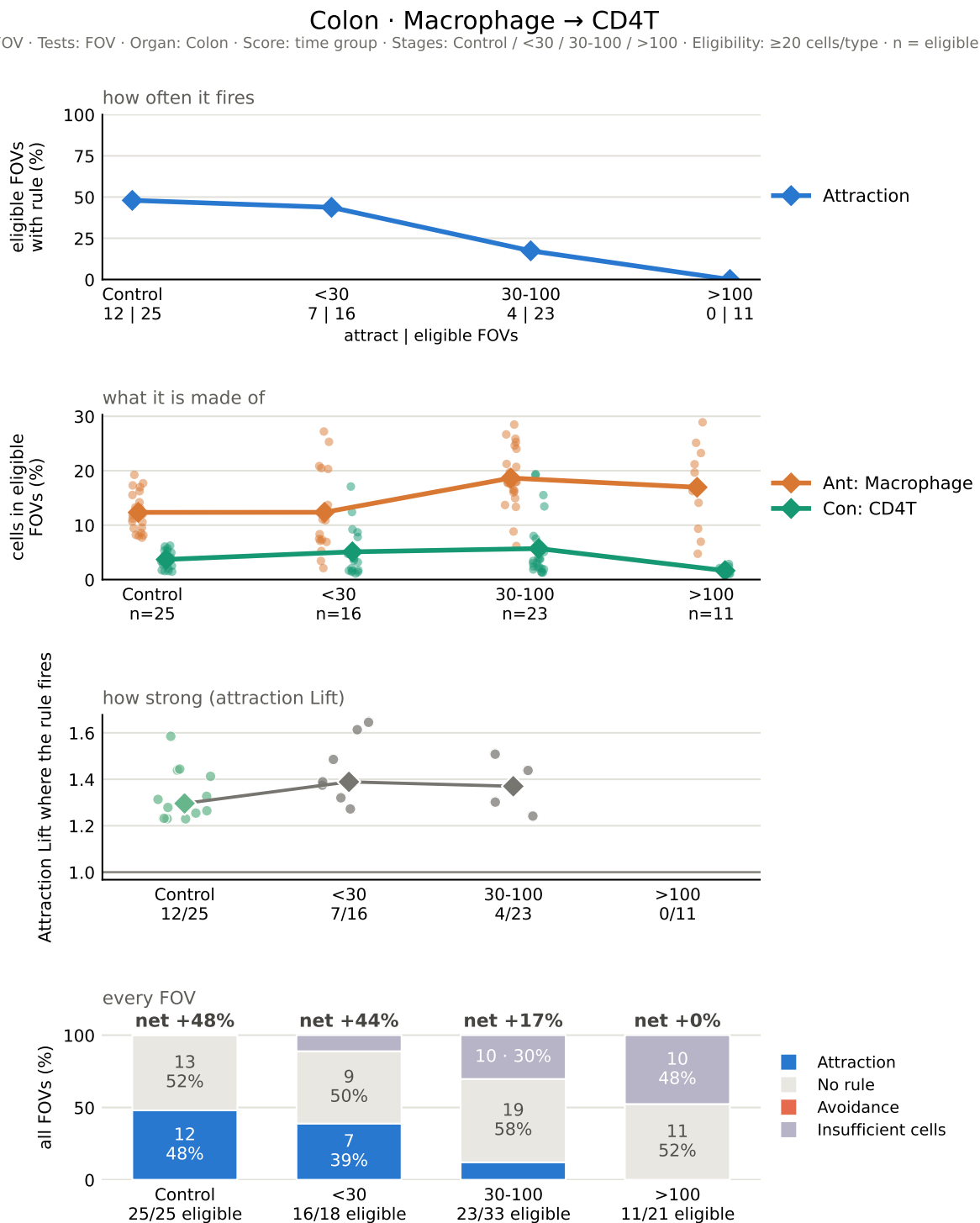


Figure 36: The four rows follow Figure 3, with time windows replacing clinical stages. Selection compares the three post-transplant windows; Control is context only.

The maps show representative attraction across the same windows.

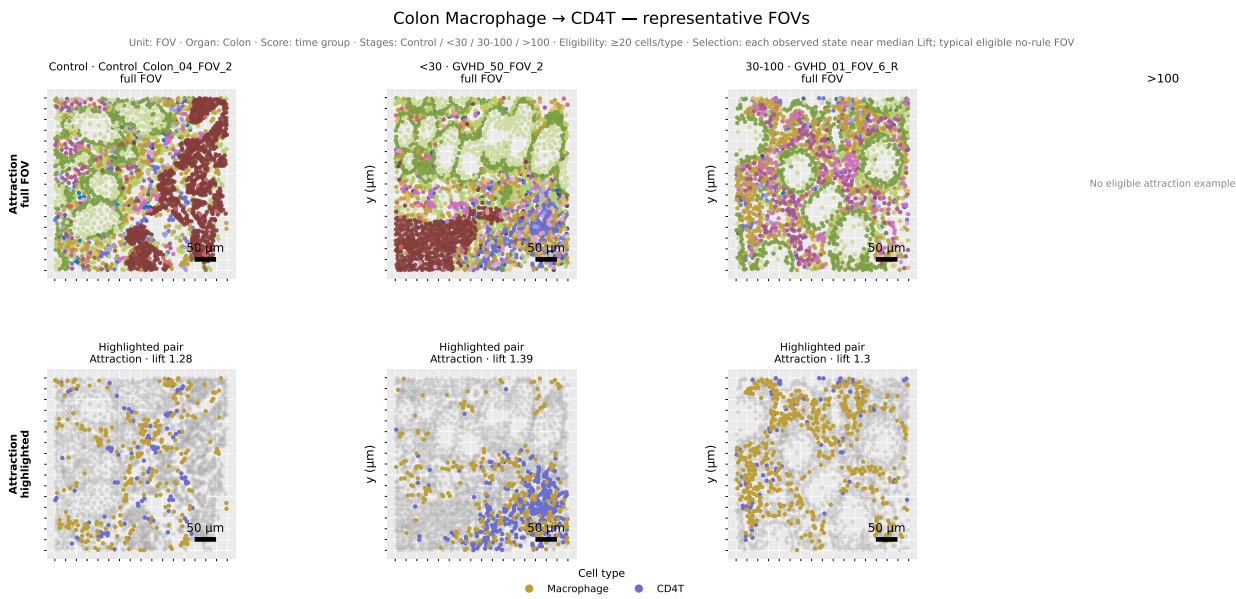


Figure 37: Maps and example selection follow Figure 17.

Eligible no-rule fields from the same windows give the contrast.

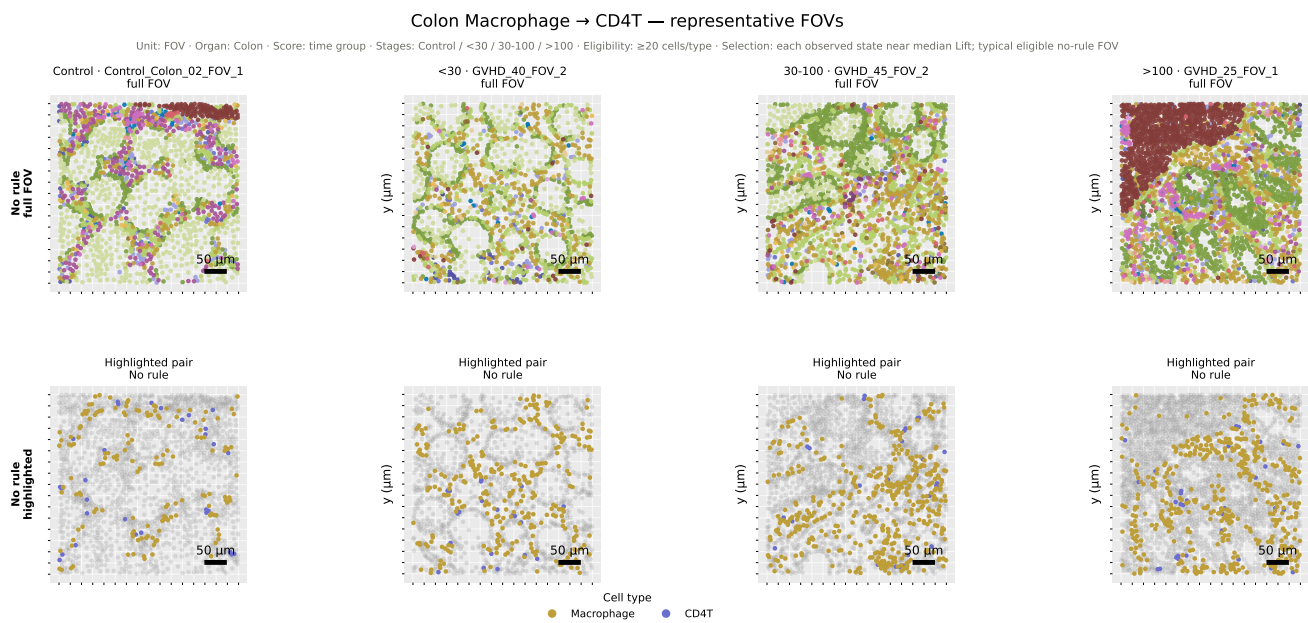


Figure 38: Maps and example selection follow Figure 17.

Duodenal Plasma_CD38 → Fibroblast attraction rises from 3/60 fields before day 30 to 11/25 after day 100, with useful denominators throughout.

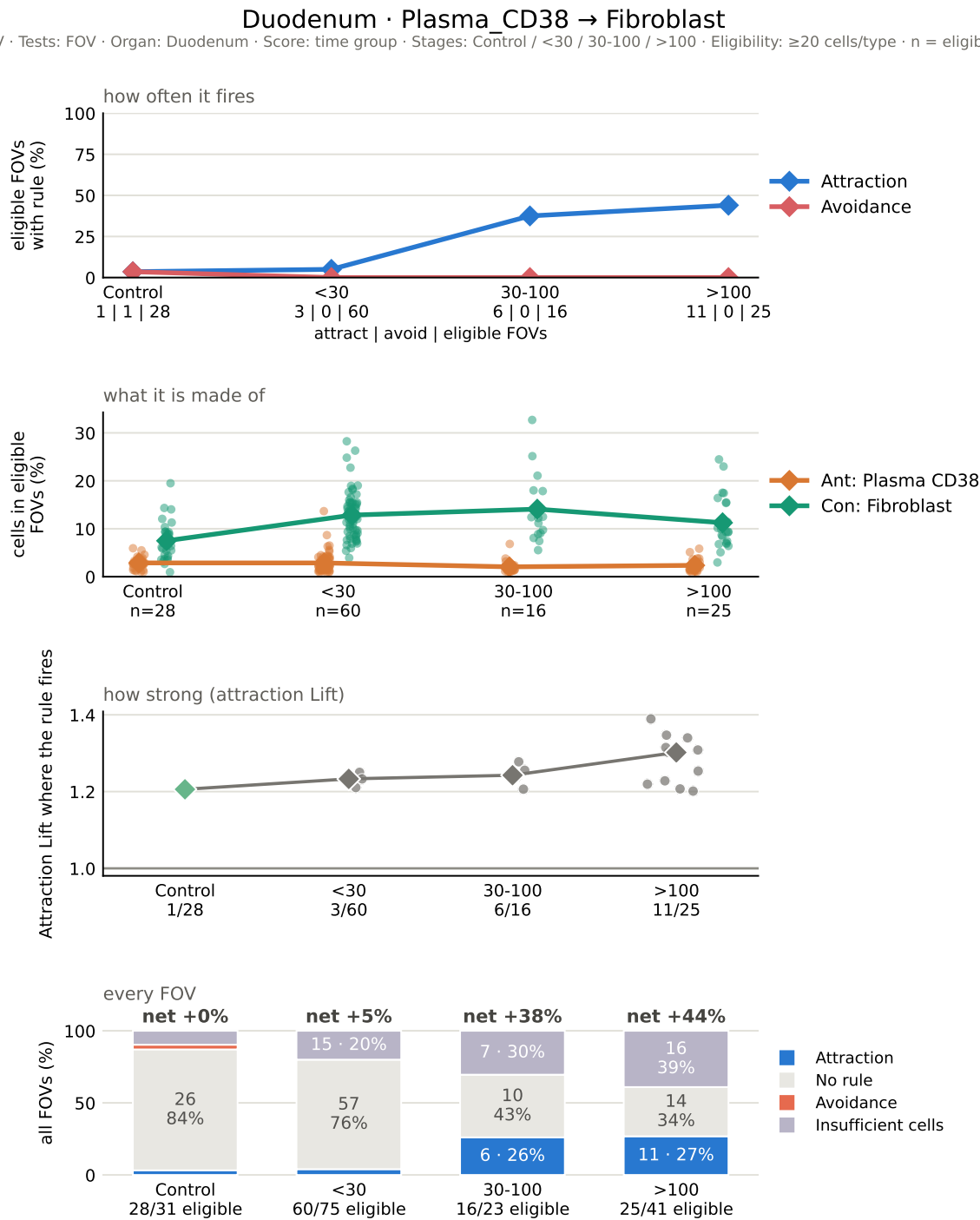


Figure 39: Read as in the preceding time figure; selection uses the 39-point post-transplant gap.

The tissue examples show the increasingly common rule.

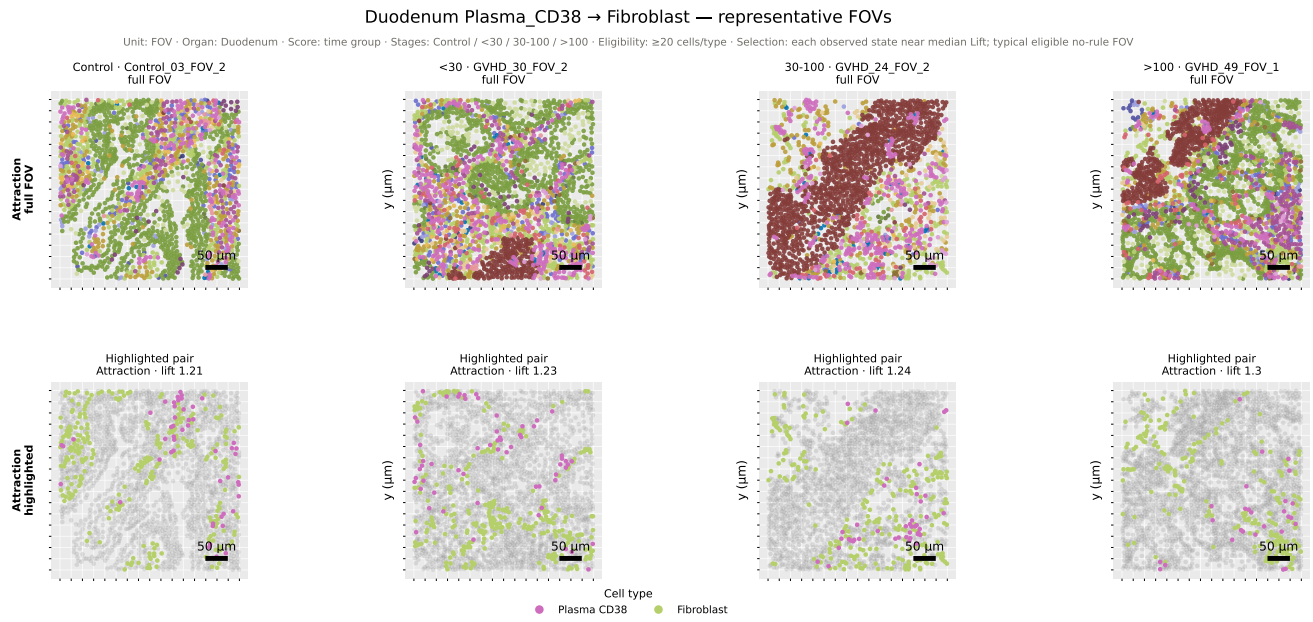


Figure 40: Maps and example selection follow Figure 17.

Eligible no-rule fields give the contrast.

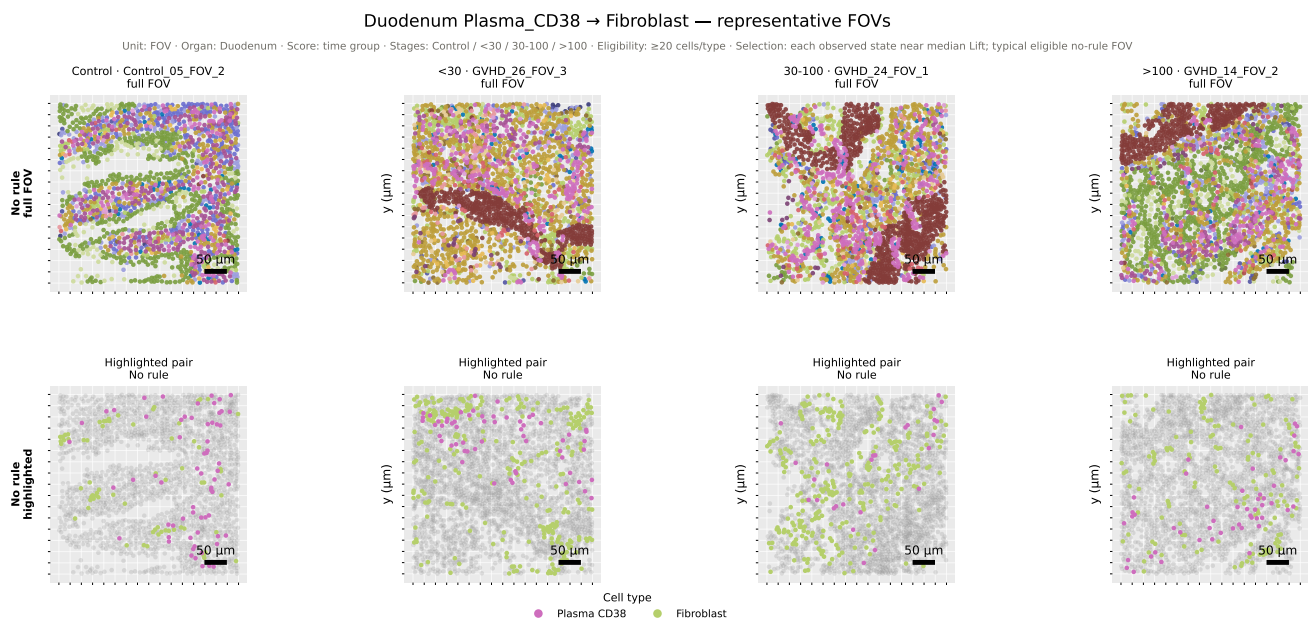


Figure 41: Maps and example selection follow Figure 17.

References

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